

Cash Management and Payment Choices Revisited

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Abstract

Exploiting unique, previously unexplored transaction-level microdata from two ECB surveys on payment and cash management attitudes of consumers in the Euro Area, I show that existing theoretical models cannot account for relevant empirical features of payment behavior. I build a dynamic model of cash management and payment choices which rationalizes two key facts: i) consumers do not choose among cash and cards based on a simple threshold-transaction size rule; ii) as long as withdrawing cash is more costly than using cards, it can be optimal to pay cashless even when cash on hand is sufficient, in order to keep cash holdings close to their optimal level. After calibrating the model to qualitatively match cross-country heterogeneity in payment and cash management patterns, I discuss possible extensions which would allow for a quantitative analysis.

Keywords: *payment choice, cash management, cashless payments, inventory models.*

JEL Codes: *E41, E42, D14.*

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1 Introduction and Overview

In this paper, I exploit unique, transaction-level data for the Euro Area and I present a novel conceptual framework to shed new light on the determinants of payment choices and cash management. Drawing upon detailed payment diaries provided in the ECB SUCH (2016) and SPACE (2019) studies, which enable me to observe consumers' *payment choice sets*, I show that existing theoretical models fail to capture some relevant dimensions of real-world payment behavior. In particular, I present evidence indicating that *transaction size policies* à la Whitesell (1989) and *cash burns policies* à la Alvarez and Lippi (2017) are not consistent with actual choices at POS. Combining payment diaries and answers to survey questionnaires, I show that i) it is crucial to jointly study cash management and payment choices; ii) cash/cashless decisions are strongly influenced both by cash holdings m and by the size s of transactions faced; iii) supply-side factors such as non-universal cashless acceptance shape cash management and payment choice policies by households.

Then, I present a novel, dynamic model of payment choice and cash management which includes these elements, combining features of Whitesell (1989) (lumpy, heterogeneous transaction sizes) and Alvarez and Lippi (2017) (payment choices that depend on cash holdings) in a unified framework. The model is built on a discrete-time version of the standard Baumol-Tobin cash management framework (Baumol (1952), Tobin (1956)) with some modifications. First, as in earlier models of payment choice (Whitesell (1989), Alvarez and Lippi (2017)) agents are endowed with a cashless payment instrument, whose usage entails a fixed cost¹. Second, agents face an exogenous, stochastic expenditure stream, in lieu of a constant, deterministic one. In any period, agents can decide whether to adjust their cash holdings by paying a fixed withdrawal/deposit cost². Afterwards, they face a random transaction size drawn from a known distribution: agents must decide whether to use cash or cashless to settle the payment, depending on current cash holdings and on the size of the transaction faced. To perform a cash payment, agents need to have enough cash on hand: as in all cash management models, holding cash entails an opportunity cost. Moreover, the possibility to use cards is not always granted, as cashless acceptance is non-universal. The uncertainty induced by the stochastic expenditure flow and by imperfect card acceptance induce a precautionary demand for cash holdings. The presence of lumpy transactions generates realistic withdrawal behavior (positive cash holdings at withdrawal, dispersion in withdrawal sizes) without the need to allow for free withdrawal opportunities,

¹This fixed cost can represent a monetary fee which is paid when using the card. It also captures other kinds of costs perceived by consumers when using cashless methods: privacy costs (as the transaction can be perceived as not being anonymous anymore) and time costs (as the literature as shown that cash payments are settled more quickly, see Klee (2008))

²This fixed cost represents both fees paid when withdrawing cash from ATMs and time costs associated with going to the bank (the famous *shoeleather costs*, as they are labeled in the monetary economics literature).

as done in [Alvarez and Lippi \(2017\)](#).

The presence of a two-dimensional state space and of stochastic, lumpy transaction sizes makes the model too complicated to obtain analytical, pencil-and-paper solutions. However, under some assumptions (which can be easily verified after solving the model numerically) I can derive results on optimal cash management and payment choice policies. The core result of this paper is presented in [Proposition 2](#), which derives features of the payment choice policy function. My model is the first to rationalize the usage of cashless payments even by consumers who have enough cash on hand to settle a transaction. I obtain a key relationship between the cost of a withdrawal and that of a cashless payment (relative to a cash transaction) which needs to hold in order to induce voluntary card usage. If paying cashless is more costly than paying by cash, but cheaper than withdrawing money from ATMs, agents may find it optimal to use their cards even when they have sufficient cash on hand, in order to keep cash holdings close to their optimal levels. In other words, in some situations cashless payments are employed as a *cash management tool*, if they are a cheaper way to control cash balances than ATM withdrawals. I show that the model's decision rules can replicate features of payment behavior which could not be explained by previous theories and that they rationalize the observed correlation between the perceived cost of withdrawals and cashless usage.

Then, I show that a calibrated version of the model can qualitatively replicate differences in payment and cash management patterns across countries. At the end of the paper, I discuss possible extensions of the model that would make it suitable for a quantitative study of payment choices and for identifying parameters using tools from the dynamic discrete choice literature.

1.1 Related literature and contribution

In recent years, innovations in financial markets and information technologies widened the set of payment methods available for individuals. Already widespread tools such as credit and debit cards became more attractive, as the birth of contactless technologies markedly improved their efficiency. As a result, the share of cashless payments has been increasing all over the world. As cash/cashless choices became of wider interest for economists, a vast literature studying the determinants of consumers' payment choices has grown.

A large part of this literature is empirical, and it made use of payment diaries or stores' transaction data. Some of the findings of this literature are extremely relevant for models of payment choice. Exploiting grocery store data, [Klee \(2008\)](#) examine the relationship between transaction sizes s and payment choices, finding that cash is mainly used for small-sized purchases, while cashless payments are prevalent when the value of the sale increases. Similar results are obtained by [Wang and Wolman \(2016\)](#) using

two billion retail transactions. Looking at the association between cash holdings m and payment choices, several papers ([Arango, Huynh, Fung, et al. \(2012\)](#), [Bouhdaoui and Bounie \(2012\)](#), [Huynh, Schmidt-Dengler, and Stix \(2014\)](#), [Bagnall et al. \(2016\)](#)) find that the likelihood of cash usage increases with the level of cash holdings. Other studies ([Arango, Huynh, and Sabetti \(2015\)](#)) use survey data to show the relevance of the probability of merchant acceptance in shaping means of payments decisions by consumers. [Huynh, Schmidt-Dengler, and Stix \(2014\)](#) and [Bagnall et al. \(2016\)](#) observe an interplay between payment choices, card acceptance and cash management decisions, finding that in economies characterized by low card acceptance rates, people hold a precautionary buffer stock of cash, seemingly in order to avoid repeated withdrawals, as they are frequently forced to pay large cash sums. In this paper, I contribute to this empirical literature in three ways. First, I provide further evidence that payment choices have an impact on cash management policies. Second, I show that payment choices by households are influenced by cash management concerns, i.e., that agents decide whether to pay with cash and cashless taking into account the effect of their choice on future cash management costs. Third, I document that expected levels of card acceptance affect both cash management patterns and payment method decisions.

There are several papers that study theoretically means of payment choices, but only a few try to jointly model cash management and payment choices by households. The two most notable attempts are those by [Whitesell \(1989\)](#) and [Alvarez and Lippi \(2017\)](#), which will be discussed at length in the main text. Both papers rationalize well-established empirical regularities. [Whitesell \(1989\)](#) derives payment choice rules that depend on the size of the transaction, showing that it is optimal for agents to use cash when $s < \bar{s}$ and cashless methods otherwise (*transaction size threshold* policies). [Alvarez and Lippi \(2017\)](#) abstract from transaction size heterogeneity and derive payment choice rules that depend on cash holdings, showing that for small payment sizes cash usage is optimal whenever agents have enough (*cash burns* policies). In this paper, I contribute to this literature in two ways. First, I exploit high-quality microdata to show that a large share of actual transactions cannot be rationalized by the two policies discussed above. Second, I combining elements of both papers in a discrete-time Baumol-Tobin framework, allowing agents to decide on how to pay depending both on cash holdings m and on the size s of the transaction. Another novelty is the explicit incorporation of imperfect cashless acceptance. This work is an attempt to bridge empirical findings that show how payment choices are multi-dimensional and the existing theoretical literature, as advocated in the final Section of [Wang and Wolman \(2016\)](#).

1.2 Relevance

The joint study of the transaction demand for cash by households and of their payment method choices is policy relevant for a variety of reasons. A first possible application is

using the model to estimate how consumers would be affected by different regulation schemes for debit/credit card interchange fees, a topic widely discussed in the IO literature ([Rochet and Tirole \(2002\)](#), [Li, McAndrews, and Wang \(2019\)](#)). Secondly, the availability of cashless payments has a first-order impact on the transaction demand for cash, with implications for the measurement of the welfare cost of inflation ([Alvarez and Lippi \(2009\)](#)). Finally, understanding the rationale behind payment choices by embedding them into models of cash management can be useful to evaluate quantitatively the welfare effects of proposed policies aimed at fighting illegal activities such as organized crime and tax evasion, implementable by banning cash ([Rogoff \(2017\)](#)) or discouraging its use ([Immordino and Russo \(2018\)](#)).

1.3 Structure of the paper

In [Section 2](#) I describe SUCH and SPACE data and I perform an empirical analysis of payment behavior in the Euro Area, highlighting some facts which cannot be rationalized by existing models of payment choice and documenting features that should be included in a more comprehensive theory. In [Section 3](#) I present my theoretical framework, I discuss properties of the solutions and I discuss how the payment choice policies that emerge from the model are a generalization of previously found ones; finally, I compare the key prediction of the model with actual data on payment choices and cash management. In [Section 4](#) I estimate the parameters of the model for all European countries in my sample, matching cross-country differences in payment and cash management behavior in the Euro Area: I discuss the results and potential extensions of the model that would be more suitable for quantitative analysis. [Section 5](#) concludes.

2 Empirical Analysis

2.1 Data description

I start by providing an overview of the two data sources used in this paper. Throughout the analysis, I exploit microdata on payment choices and cash management behavior of individuals from 15 Euro Area countries, combining two datasets provided by the ECB jointly with two publications: the first one, released in 2016, is the Study on the Use of Cash by Households (SUCH from now on); the second one, released in 2020, is the Study on the Payment Attitudes of Consumers in the Euro Area (SPACE from now on)³. Each study contains a payment diary and a survey questionnaire: in the diary, participants recorded information on all their payments and adjustments of cash bal-

³For further information about the methodology and results of these two studies, see respectively [Esselink and Hernández \(2017\)](#) and [ECB \(2020\)](#).

TABLE 1: Summary statistics.

	Mean	Median	SD	Min	Max	<i>N</i>
<i>Payment choices</i>						
Payment size (€)	21.38	8.52	87.35	0.01	16,000.00	155,565
Paid cashless	0.26	0.00	0.44	0.00	1.00	155,565
Cashless accepted	0.74	1.00	0.44	0.00	1.00	152,360
Cashless possible	0.68	1.00	0.47	0.00	1.00	129,158
Cash possible	0.89	1.00	0.31	0.00	1.00	178,472
Both methods possible	0.53	1.00	0.50	0.00	1.00	129,169
Paid cashless Both poss.	0.32	0.00	0.47	0.00	1.00	69,013
<i>Cash management choices</i>						
Cash holdings (€)	68.47	35.77	152.86	0.00	9,990.92	178,472
Withdrawn	0.03	0.00	0.18	0.00	1.00	90,843
Withdrawal size (€)	91.93	50.00	161.11	0.45	3,600.00	4,063
Adjusted	0.08	0.00	0.27	0.00	1.00	90,843
Adjustment size (€)	75.03	35.00	145.01	0.05	3,600.00	6,502

Note: Adjustments are defined as any increase in cash holdings between a payment and another; they also include receiving money from friends, or as cash income. Withdrawals are a particular type of adjustment: in these cases, agents either withdrew money from an ATM or got it from a bank teller. *Source: ECB SUCH (2016) and SPACE (2019) Data.*

ances in a given day; within the questionnaire, they were asked a number of questions on their preferences about payment methods, on the set of cashless instruments they had access to and on their cash management behavior.

The main variables are summarized in Table 1. A key characteristic of these data is that one can combine the information provided to track down the time path of cash holdings and to elicit *payment choice sets* of consumers (whether cash or cashless or both could be used to carry out purchases) for each transaction during the day of analysis. This feature enables me to distinguish between *voluntary payments* and *forced* ones. I define a payment as *voluntary* when the agent has both options available: she holds sufficient cash, she has access to cashless payment method and the current store accepts cashless payments. Payment method choices that do not satisfy one of these conditions are *forced*, as in these situations the payment choice set collapses to a singleton ⁴.

The ability of observe cash holdings at each point in time creates a unique setup to study payment choices: in previous studies that analyzed payment behavior, payment

⁴Appendix A.1 contains details on the data cleaning procedure that enables me to derive the time path of cash holdings and therefore to elicit payment choice sets. In all the analysis, I abstract from imperfect cash acceptance. It is now quite common for stores to avoid accepting cash from clients. In SUCH data there is no information on this; in SPACE data agents were asked about cash being accepted, but the extent of recorded non-acceptance in the data is so tiny that I do not consider it explicitly.

choice sets are not appropriately taken into account because cash holdings are typically unobserved. [Klee \(2008\)](#) exploits grocery store data, thereby ruling out imperfect acceptance of cashless payments: however, her strategy does not differentiate between voluntary cashless payments (in which the agent also had the opportunity to pay with cash) and forced ones (in which the agent did not have enough cash to carry out the purchase). A similar problem is encountered by [Wakamori and Welte \(2017\)](#): while they do observe perceived payment acceptance by agents, they don't know how much cash do they have at each point in time. In both papers, the authors can *limit* the choice sets of agents, but do not *observe* the choice set in an exact way. The data used in this paper makes choice sets exactly observable: this allows me to understand if a choice is driven by preferences or it is just the by-product of a trivial payment choice set.

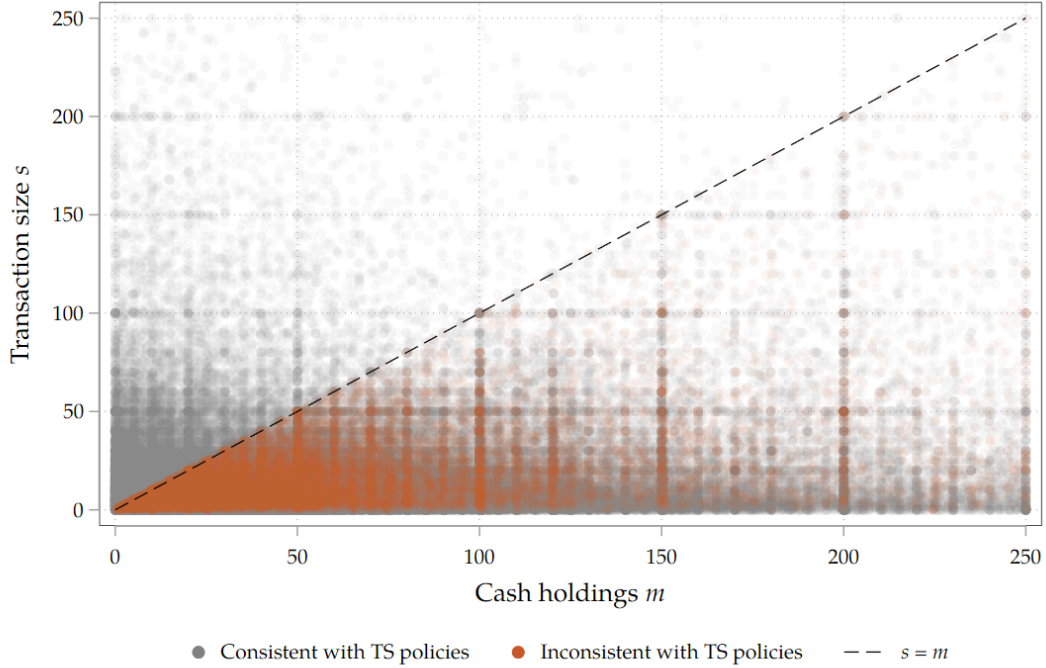
2.2 Payment choice models and observed payment behavior

The main goal of this Section is to re-examine the predictions of existing theoretical models of payment choice, to see if their results can rationalize payment behavior we see in the data. To this end, the ability to exactly observe payment choice sets is crucial: only voluntary payments are informative on agents' decision rules. Forced cash and cashless transactions, instead, cannot inform us on the determinants of payment choices.

Before analyzing whether existing models of payment choice are able to rationalize observed behavior, I need to briefly describe the key features of these frameworks. Most models of payment choice have a common element: they usually are enhanced versions of the standard cash management model by [Baumol \(1952\)](#) and [Tobin \(1956\)](#). In this basic model, agents must finance an exogenous, deterministic expenditure stream using cash. They want to avoid holding too much cash on hand, since it has a flow opportunity cost; at the same time, they dislike holding too little money, as withdrawals entails a fixed cost. This gives rise to a problem whose solution is the famous *square-root rule*, a standard formula in cash management theory that yields the optimal withdrawal: agents withdraw m^* and they wait their cash balances to reach zero; then, the cycle starts again. As payment method choices and cash management decisions are strictly related and jointly determined (I will provide evidence about this in [Subsection 2.3](#)), the most natural way to model payments is to embed a cash/cashless choice into a cash management framework similar to the one just described.

The first paper to do so is [Whitesell \(1989\)](#), where a static Baumol-Tobin model is augmented with a distribution of transaction sizes and a choice among cash and cashless payments. He finds that it is optimal to pay by cash whenever the size of the transaction is smaller than a given threshold $\bar{s}$, and to pay using cards otherwise. These decision rules (called *transaction size threshold* policies) seem to be consistent with the widely observed phenomenon that people settle larger transactions using cashless payments

FIGURE 1: Whitesell (1989)'s transaction-size threshold policies and the data.

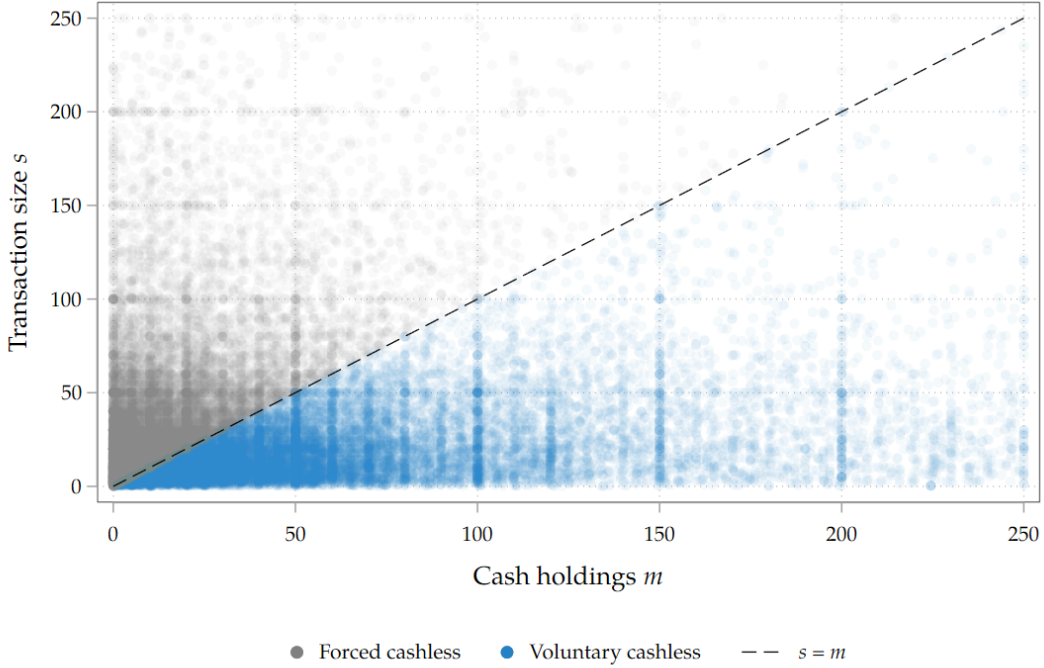


Note: Each dot is a transaction. Obs. flagged in orange are not coherent with TS policies, i.e., either i) the agent voluntarily used cash to settle a transaction bigger than her largest voluntary cashless payment, or ii) the agent voluntarily used a cashless payment method to settle a transaction smaller than her smallest voluntary cash payment. Source: ECB SUCH (2016) and SPACE (2019) Data.

than they do using cash, a behavior observed both in early studies on payment choices (Boeschoten and Fase (1989), Boeschoten (1992)) and in more recent papers that made use of high quality microdata (Klee (2008), Bagnall et al. (2016), Wakamori and Welte (2017), Arango-Arango et al. (2018)). It is possible to test the predictions of Whitesell (1989) using SUCH/SPACE data, where transaction sizes and payment methods chosen are directly observed. Since I observe more than one payment (sometimes as much as eight) for a large share of people, it is possible to test transaction size threshold policies even allowing the threshold $\bar{s}$ to be heterogeneous. If thresholds are individual-specific and given by $\bar{s}_i$, according to Whitesell (1989) one should never observe an individual performing a voluntary cash payment which is larger than her biggest voluntary cashless payment. Figure 1 shows that a large share of people explicitly violate transaction size threshold policies. Given that payment diaries in the data are only one day long, one can expect to see much more agents violating such rules as the observation time horizon is extended.

In the dynamic framework of Alvarez and Lippi (2017), instead, payment method choices depend only on cash holdings m , with the size of the transaction not really playing any role, as the authors rely on an infinitesimal and deterministic expenditure stream. Their model's prediction is that agents should use cash whenever they have

FIGURE 2: Payment method choices, individual transactions.



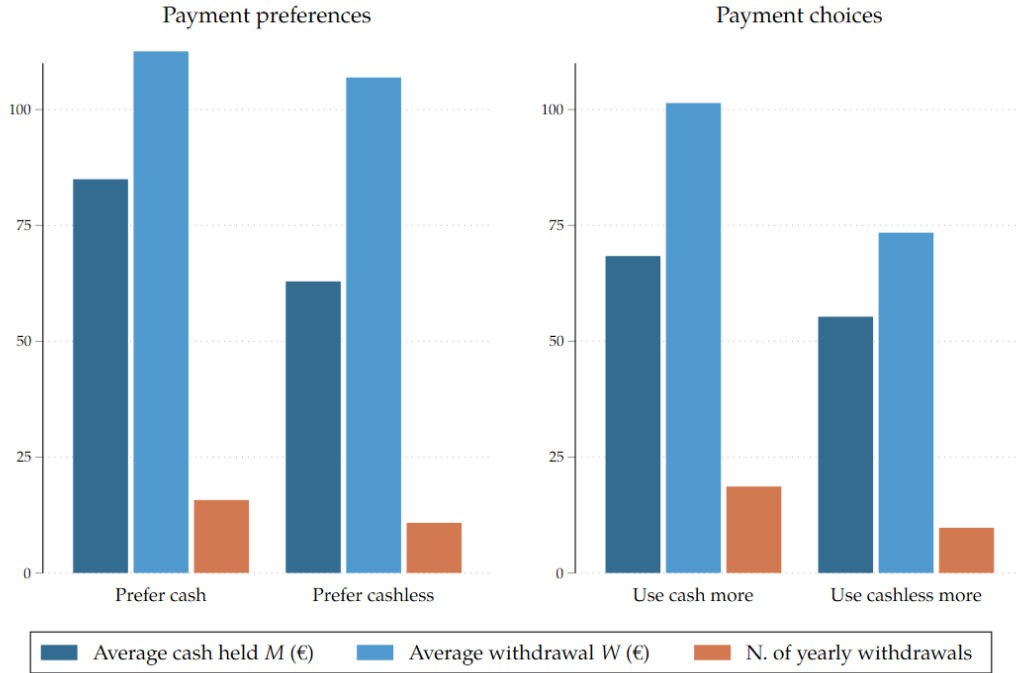
Note: Each dot is a transaction reported in payment diaries. Above the 45 degree line, all payments must be settled using cards ($m < s$). The blue dots below the 45 degree line represent voluntary cashless payments, i.e., transactions where cashless payments were employed even if the agent had sufficient cash to pay. Source: ECB SUCH (2016) and SPACE (2019) Data.

some on hand: these decision rules (called *cash burns* policies) are consistent with empirical evidence showing that the likelihood of using cash increases with the level of cash holdings (Arango, Huynh, Fung, et al. (2012), Bouhdaoui and Bounie (2012)). Despite being quite accurate for very small payments, Figure 2 shows that *cash burns* policies cannot rationalize a large fraction of transactions which are bigger in size⁵, for which agents often employ cashless methods even if they have enough cash on hand.

There is no model based on explicit individual optimization that rationalizes voluntary cashless payments without resorting to simple threshold size policies that are contradicted by the data. Producing such a model, and studying its theoretical properties,

⁵As Alvarez and Lippi (2017) directly point out, their results are only applicable for very small transaction amounts, as they are obtained from a model where the expenditure stream is infinitesimal. Therefore, it should not be surprising that cash burn policies do not match the data when $s \gg 0$: the authors themselves acknowledge that their explanation is “complementary to that based on transaction size” (as in Whitesell (1989)) and “only able to address the choice of means of payment for small-sized transactions”. Despite this, cash burns policies have been employed even in models with heterogeneous transaction sizes, without proving their optimality in this new setting. For instance, Arango, Bouhdaoui, et al. (2014) assume that agents follow a generalized cash burn policy: they use cash whenever they have enough, but use their card when they incur in a transaction which is too big. Even in their model, agents never use cards when they have enough cash to carry out a payment, contradicting the data. In this paper, I show that generalized cash burn policies are not optimal in a model with heterogeneous transaction sizes.

FIGURE 3: Payment preferences, payment choices and cash management policies.



Note: This graph displays cash management policies across groups of individuals defined according to their payment preferences (on the left) and to their payment behavior in the day of the diary (on the right). *Source: ECB SUCH (2016) and SPACE (2019) Data.*

is the aim of this work. I now exploit the information provided by SUCH and SPACE data in order to understand which are the relevant features of payment behavior that a model should necessarily include.

2.3 Payment behavior: some key facts

I will now describe some features of payment behavior that can be observed using SUCH and SPACE data⁶. These stylized facts will be important to understand the basic elements that a model of payment choice should include. In particular, I argue that i) as done by previous models, jointly modeling payment and cash management decisions is crucial, and ii) a realistic model of payment choice has to feature decisions rules that depend both on cash holdings m and on the size of transactions faced s . I also provide evidence showing that imperfect cashless acceptance is a key determinant of payment choice and cash management patterns.

⁶The goal of this Section is not to give a complete overview of payment and cash management behavior in the Euro Area, but to highlight key relationships that should drive the modeling of payment choices. See respectively [Esselink and Hernández \(2017\)](#) and [ECB \(2020\)](#) for a descriptive analysis of payment and cash management patterns in 2016 and 2019.

2.3.1 The relationship with cash management

As the previous literature extensively discussed⁷, there is a mutual relationship between payment and cash management choices: on one hand, people might manage their cash in a different way based on how much they use cashless payments; at the same time, when choosing on how to pay people might take into account the impact of their decision on future cash holdings.

Let's first look at how payment preferences and choices impact cash management strategies. Figure 3 provides further evidence on the existence of this channel. In the left panel of the Figure, I look at cash management behavior across groups defined in terms of expressed payment preferences, which agents are asked about in the survey questionnaire and which are coherent with observed choices in the diary, as will be shown later on. It is possible to see that people who reportedly prefer to pay with cash will have higher average cash holdings, withdraw more often and perform larger withdrawals than people who state they prefer to pay cashless. In the right panel, I divide people in groups based on the payment method they used more often during the day of the diary and I obtain similar results. These plots suggest that people choose their cash management strategy taking into account their payment choice habits/preferences.

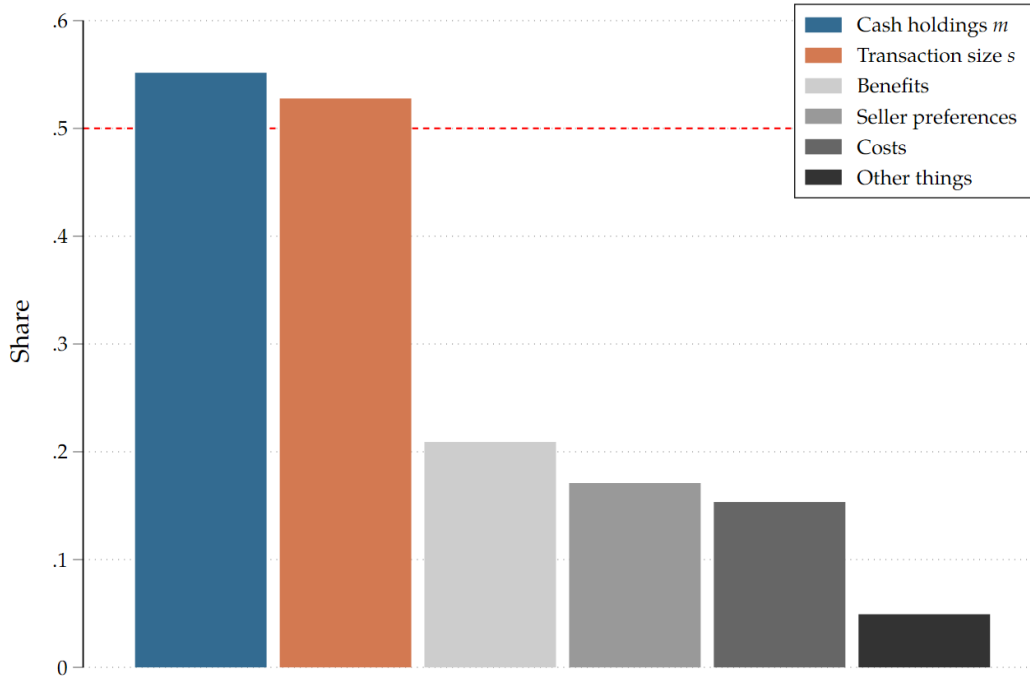
Key Fact 1. *People that reportedly prefer or that mostly employ cash for payments used to carry more cash with them, to withdraw bigger amounts and to do so more often.*

At the same time, people could take into account expected cash management costs when deciding on how to pay, if they have both options (cash and cashless) available. To see this, first notice that from a cash management perspective, the two payment methods are inherently different: when holding m units of cash, paying a transaction of size s using a cashless instrument leaves cash balances unchanged (future cash balances m' are equal to m), whereas using cash depletes them ($m' = m - s$). When facing an opportunity to pay with both methods, it is natural to think that agents will weigh the static cost/benefit of performing a cashless payment⁸ against the dynamic gain/loss associated with having higher future cash balances: since the dynamic component will depend on both m and s ($m' = m - s$ in case of a cash payment), it makes sense that these are important determinants of their choice. Fortunately, in the SUCH questionnaire people were directly asked about this, so I can test if they really care about m and s when making a payment method decision. I do so in Figure 4, where I plot the shares of agents that say if certain factors (among which m and s) affect their means of payments choice. We see that a large share of agents (more than 50%) takes

⁷There is a large body of papers making the claim that cash management and payment choices should be studied jointly as they mutually affect each other. Applied papers making this claim are Bagnall et al. (2016) and Wang and Wolman (2016). As said in the previous Section, theoretical payment choice models usually feature a cash management problem.

⁸It will be a cost for people that prefer cash as a means of payment, and a benefit for people that like more to settle their transactions using cards.

FIGURE 4: Determinants of consumers' choice of payment instrument at the point of sale.



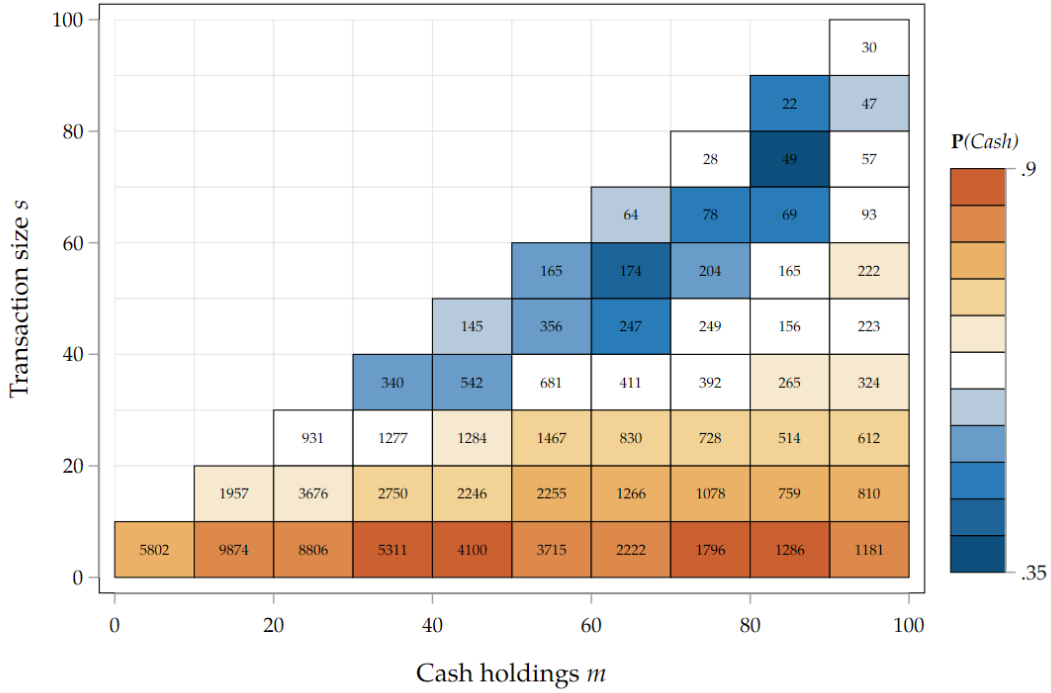
Note: This graph displays the shares of people reporting that a certain factor influences their payment decision. The question respondents answered was: “Which of the following influences your decision to pay with cash or card or other non-cash payment methods?”. Multiple responses are possible. Source: ECB SUCH (2016) Data.

into account these two factors when paying. We also see that compared to other factors, the size of the withdrawal and the size of the transaction are perceived as much more crucial for choosing on how to pay.

Key Fact 2. *The most relevant factors in determining the choice of a payment method are cash holdings m and the size of the transaction s .*

The fact that both m and s matter for means of payments decisions (which is consistent with payment choices being influenced by cash management concerns) is confirmed by data on actual transactions. In Figure 5 I display the share of transactions settled in cash for different values of m and s , in situations where both payment methods were accessible to consumers. Several interesting patterns emerge. First, it seems that for any given level of s , agents tend to use cash more when m is larger. Second, keeping m fixed, it seems that agents tend to use cash more for low values of s (where cash usage predominates) than for high values. Finally, when s is very close to m (on the diagonal) cashless is employed much more often. This seem to suggest that agents want to avoid using cash when this leads to a depletion of their money holdings, if they can avoid doing so. This effect is more pronounced when cash holdings are at an intermediate level (around 40-60 euros) with respect to situations where cash balances are

FIGURE 5: Share of cash payments for different m and s .



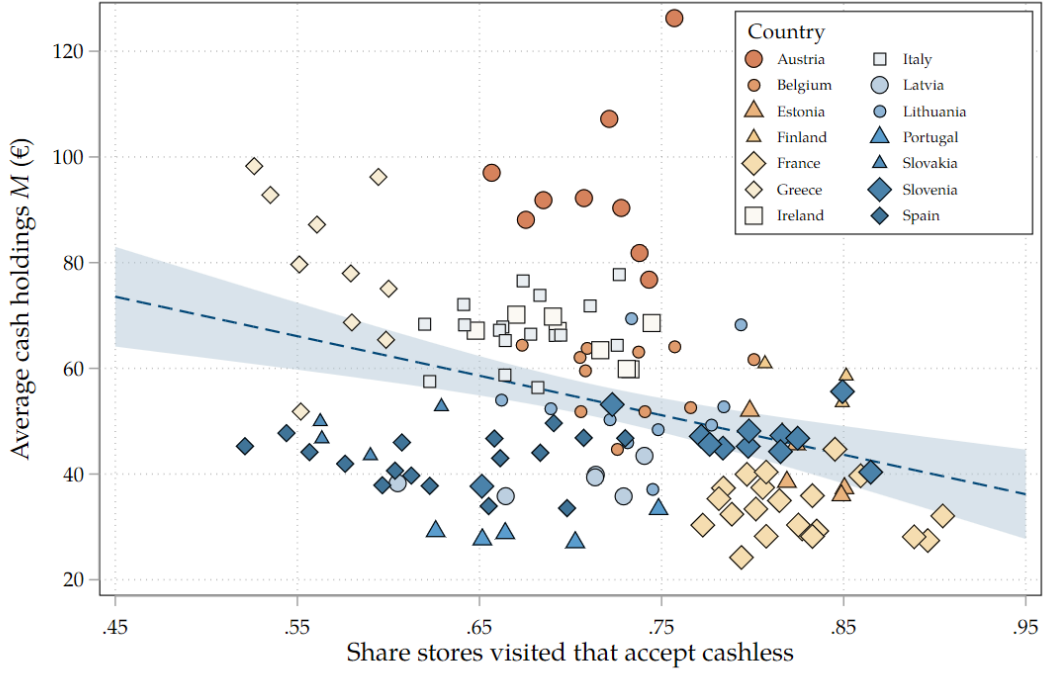
Note: This graph displays the shares of people paying with cash (when both payment options are possible) for bins defined in terms of cash holdings at payment (m) and transaction size faced (s). The share of people paying by cash is represented by the color of each cell, whereas the numbers displayed on cell denote the number of observations falling in that particular (m, s) bin. I focus on transactions where m and s are smaller or equal than 100 euros to avoid having cells with a very small number of observations. Source: ECB SUCH (2016) and SPACE (2019) Data.

low. These features of the data are consistent with the theory that payment choices are, at least to some extent, driven by the desire to minimize expected cash management costs: if agents desired to hold some optimal level of cash holdings m^* , they will be more inclined to pay by cash when the cash balances are far from m^* (for instance, if they are very low) than when they are close to it.

Key Fact 3. *The probability of paying by cash is increasing with respect to cash holdings m and decreasing with respect to the size of the transaction s .*

The fact that both m and s are key determinants of payment method decisions offers a possible explanation for why the models by Whitesell (1989) and Alvarez and Lippi (2017) are unable to rationalize a large portion of the data: both frameworks do not feature decision rules that depend on both variables. In Whitesell (1989), as the model is static, agents' decisions are taken at the start of the period: there is way of keeping track of cash balances, and choices do not depend on cash on hand m . In a completely opposite fashion, Alvarez and Lippi (2017) do not take into account heterogeneity in transaction sizes s and their effects on payment choices. In the model I will present in Section 3, dependence of means of payment choice on m and s will be explicitly

FIGURE 6: Average cash holdings and cashless acceptance rates in different Euro Area NUTS 2 regions.



Note: This graph plots, for each Euro Area region (NUTS 2 classification), average cash holdings in that region against the regional share of shops that accept cashless payments. A linear fit with 95% confidence intervals is overlaid to the graph. *Source: ECB SUCH (2016) Data.*

modeled in a dynamic cash management framework.

2.3.2 The role of card acceptance

Most existing models of payment choices and cash management do not take into account the fact that cashless methods are not universally accepted.⁹ This omission could lead to theoretical results which are not coherent with observed behavior, for two reasons: i) imperfect cashless acceptance introduces a precautionary motive for holding cash; ii) payment choices are influenced by card acceptance levels. I now provide evidence on these mechanisms.

First, I consider the association between imperfect cashless acceptance and the level of cash holdings. If cards are not universally accepted as a means of payment, consumers know that they may encounter a situation in which i) they don't have enough cash with them to settle a transaction; ii) the store does not accept their cashless payment method. One would expect that, in response to this, agents will optimally hold more

⁹In [Alvarez and Lippi \(2017\)](#), the authors explicitly acknowledge the importance of cashless acceptance (and including transaction sizes too) for modeling cash/cashless choices, saying that "future models might benefit by unifying those aspects into a single model and quantify the relative importance of each of these frictions by using the relevant micro data".

TABLE 2: Estimation results for model (1).

	<i>Dependent variable: Paid cash</i>			
	(1)	(2)	(3)	(4)
Cashless accept. (%)	-0.0095*** (0.00022)	-0.0035*** (0.00074)	-0.0010 (0.0011)	0.0030* (0.0012)
Cash holdings m	0.094*** (0.0019)	0.076*** (0.0019)	0.020*** (0.0027)	0.020*** (0.0027)
Transaction size s	-0.14*** (0.0013)	-0.13*** (0.0014)	-0.093*** (0.0023)	-0.093*** (0.0023)
Withdrawn before payment	0.38*** (0.0048)	0.37*** (0.0063)	0.35*** (0.0082)	0.35*** (0.0081)
Prefer cashless		-0.24*** (0.0044)	-0.28*** (0.0058)	0.14** (0.054)
No preference		-0.10*** (0.0047)	-0.11*** (0.0061)	0.17** (0.061)
Prefer cashless $\times$ Cashless accept. (%)				-0.0056*** (0.00071)
No preference $\times$ Cashless accept. (%)				-0.0037*** (0.00081)
Observations	53987	47677	28751	28751
Robust SE	✓	✓	✓	✓
Controls		✓	✓	✓
Only voluntary payments			✓	✓
Interaction				✓

Note: In addition to controls shown, I also include demographics (sex, age, education, employment status, income, household size, internet usage), transaction-specific information (day of the week, month, type of purchase/store), region, population density (urban/rural) and year fixed effects. I only use 2019 data as 2016 data does not contain the urban/rural information, which is crucial for acceptance rates. *Source: SPACE (2019) Data.*

cash in order to relax their cash-in-advance constraints. In other words, where cashless acceptance rates are lower, there is a stronger *precautionary motive* for holding cash. [Figure 6](#) suggests that this mechanism is actually in place. In the Figure, I plot average cash holdings and cashless acceptance rates for each NUTS 2 level region in SUCH data. The plot shows that in regions with high cashless acceptance rates, agents hold less cash. This seem to confirm that there exist a precautionary motive for holding cash related to uncertainty over future acceptance of alternative payment methods.

Key Fact 4. *In areas with higher cashless acceptance rates, agents hold less cash.*

Perceived rates of acceptance of cashless instruments may also affect payment behavior. Agents could decide on how to pay based on acceptance rates they expect to encounter in future payments. To see why, think of two consumers, A and N , both endowed with a cashless instrument (say a debit card). Both prefer to settle their purchases in

cash, but still use the card when they run out of cash. *A* lives in an area where card acceptance is universal; *N* lives in an area where a large share of shops do not accept her payment card. Suppose that both consumers face a transaction in which they can use whichever payment instrument they like: given that holding cash has a higher precautionary value for *N* (she is not sure of being able to use the card if she runs out of cash), we expect her to use cashless payments more frequently in such circumstances in order to preserve cash for her incoming payments. I test this hypothesis by estimating the linear probability model

$$\begin{aligned} \Pr(Cash_{ijt} = 1) = & \beta_0 + \beta_1 AccRate_j + \beta_2 PrefMethod_i + \\ & + \beta_3 (AccRate_j \times PrefMethod_i) + \\ & + \beta_4 m_{it} + \beta_5 s_{it} + \beta_6 W_{it} + \gamma \mathbf{X}_{it} + \eta \mathbf{D}_i + \varepsilon_{it}, \end{aligned} \quad (1)$$

where $AccRate_j$ is the share of stores accepting cashless payments in the geographical area j , identified by the combination of a NUTS-2 level region and a urban/rural dummy. I control for key determinants of choices at POS, such as cash holdings m_{it} , transaction size s_{it} and a dummy equal to one if the agent has withdrawn cash right before the payment (W_{it}). Other demographic and transaction-specific controls are also included. Estimation results are displayed in [Table 2](#). When no controls are included and all the observations are taken into account, the relationship between cashless acceptance and the likelihood of a cash payment seems strongly negative. Part of this effect, however, is due to selection: controlling for preferences (it is more likely that people prefer cashless, and therefore use it, in regions with high card acceptance) the negative effect shrinks by two thirds in magnitude. By focusing only on voluntary payments, where both payment methods were available to the consumer (as it is much more likely that cashless was not even an option in region with low card acceptance rates), the observed association between acceptance rates and payment choices is no longer significant. Breaking down the relationship across preference groups, we see that for people who prefer cash (the baseline group), lower levels of card acceptance are associated with smaller probabilities of using cash. In particular, a 10-percent decrease in the share of stores that accept cashless methods in the area where the agent lives is associated with a decrease in cash usage (when both options are available) of around 3%.

Key Fact 5. *In areas with lower cashless acceptance rates, agents are more likely to use cards when they are offered the chance to do so.*

3 Theoretical Model

Why a model? A natural question to pose is why do I need a formal theoretical model of payment choice. The core idea is that, after showing features of the data which are not

included in previous theories and after illustrating that many observed patterns are not explained by previous models, the goal of this paper is to illustrate theoretically that such payment patterns can be the consequence of rational choice in a dynamic cash management setting. The model will enable us to understand what are the conditions under which voluntary cashless payments are optimal, and to provide a theoretical backing for the widespread observation that smaller payments are usually settled with cash. Another benefit of laying down a formal theory is that it can be used to evaluate the effect on payment and cash management behavior of structural changes that map to model parameters, such as increases in ATM densities, improvements in cashless acceptance and so on. These are questions that cannot be answered without a structural model of behavior.

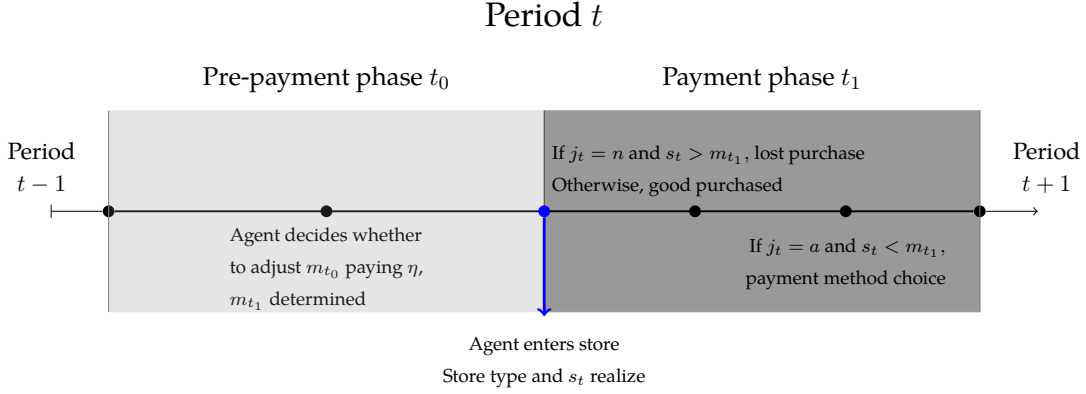
In a nutshell. Consider the problem of an agent who has access to two types of payment instruments: cash and a cashless payment method (e.g., a debit card)¹⁰. There are (potentially) three differences between cash and cashless payments: i) the cost of paying using cash or using a cashless instrument may be different; ii) the use of cashless instruments does not deplete cash balances, whereas paying by cash obviously does; iii) cash is always accepted in stores, whereas cashless instruments may not be accepted. Both withdrawing and holding cash are costly. The only way to obtain cash is to withdraw it from an ATM. Each period, agents exogenously receive the need to purchase a good or service of a certain value: the transaction can be settled either through cash or (in some stores) using some cashless method. Purchasing items yields an utility value: for this reason, a lost purchase (i.e., being unable to buy the desired good, which can occur if the agent has insufficient cash and the store she is in does not accept cards) entails an utility loss. The agent also needs to take into account the costs related to cash management and carrying out payments. She will choose the optimal amount of money to hold at each point in time, as well as the way in which she wants to settle each transaction. The underlying cash management setup is similar to the standard problem by [Baumol \(1952\)](#) and [Tobin \(1956\)](#); I add elements of payment choice models such as [Whitesell \(1989\)](#) (lumpy, heterogeneous and stochastic transaction sizes) and [Alvarez and Lippi \(2017\)](#) (dynamic setup with choices that depend on money holdings). Another novelty of the model is the explicit inclusion of imperfect acceptance of cashless payments, which was ignored in previous frameworks, and which leads to a precautionary motive for holding cash in order to avoid lost purchases.

3.1 Setup

The timing of the model can be visualized in [Figure 7](#). Time is discrete and indexed by t . Each period t represents a payment, and it is divided in a pre-payment phase t_0 and

¹⁰Throughout the paper and in the model, I never deal with the issue of card ownership. I decided to omit this margin as around 98% of people in SUCH and SPACE data have access to at least one cashless means of payment, making card ownership a second order issue for payment choices.

FIGURE 7: Timing of the model.



in a payment phase t_1 ¹¹. The agent discount the future at rate $\beta \in (0, 1)$.

In the *pre-payment phase*, agents decide whether to adjust their money holdings m_{t_0} by paying a fixed cost η or to keep them unchanged. In the *payment phase*, they enter a store where expenditure size s_t (which is a continuous random variable with probability density function $f(s)$ and cumulative distribution function $F(s)$) realizes. Purchasing a good of size s_t delivers utility $u(s_t)$. Stores are of two types $j \in \{a, n\}$, with $\Pr(a) = \phi$: stores of type a accept cashless payments, while stores of type n don't. I assume that all agents own a cashless payment instrument and that they discover the size of the purchase s_t and store type j_t simultaneously, after entering the store. Let m_{t_1} be cash holdings in the payment phase of period t . In stores that accept cashless payments, the agent will be free to choose payment method whenever $s_t \leq m_{t_1}$; if $s_t > m_{t_1}$, she will clearly use cashless payments. In stores that do not accept cashless payments, the agent will pay in cash whenever $s_t \leq m_{t_1}$; when $s_t > m_{t_1}$, she will not be able to complete the transaction and purchase the good/service, losing the associated utility.

I assume that cashless payments entail a fixed cost/benefit κ with respect to cash ones: the parameter κ measures taste for cash or cashless payments. As it is standard in the cash management literature, I also assume that bearing cash is costly for individuals: holding m units of cash in the payment phase entails a variable cost Rm . Moreover, I assume that $m \geq 0$ at each point in time: agents cannot have negative cash holdings.

Agent's problem. Given an initial condition m_{0_0} , the aim of the agent is to solve the discounted utility maximization problem

$$\max_{\{m_{t_0}, m_{t_1}, \tau_i, \lambda_i\}_{t=0, i=0}^{\infty}} \mathbb{E}_0 \left(\sum_{t=0}^{\infty} \beta^t (u(s_t) - Rm_{t_1}) - \sum_{i=1}^{\infty} (\beta^{\tau_i} \eta - \beta^{\lambda_i} \kappa - \beta^{\gamma_i} s_{\tau_i}) \mid m_{0_0} \right),$$

¹¹The setup is similar to [Arango, Bouhdaoui, et al. \(2014\)](#), both concerning the division of each period in two sub-periods and the presence of discrete-sized expenditures. However, there is a major difference: in their model, the size of cash adjustments are not endogenous but randomly drawn from the observed distribution, and the authors assume the optimal payment policy is a *cash burns* policy (as in [Alvarez and Lippi \(2017\)](#)) with a minimum cash-holding threshold. In this model, there is explicit optimization by households.

where $\{\tau_i\}_{i=1}^\infty$, $\{\lambda_i\}_{i=1}^\infty$ and $\{\gamma_i\}_{i=1}^\infty$ are respectively times of cash adjustments, cashless payments and lost purchases. The agent needs to decide the timing and size of adjustments and which payment method to use in each occasion in which she has the choice. I now illustrate the problem in simpler, dynamic programming notation.

Dynamic programming problem. Let $\hat{V}(m)$ denote the value of the problem for an agent with cash holdings m in the pre-payment phase.

$$V(m) = \max \left\{ \max_{\hat{m}} EV(\hat{m}) - \eta, EV(m) \right\}. \quad (2)$$

The interpretation is straightforward: agents can decide whether to pay the fixed cost η to go to an ATM and adjust their money holdings (either by withdrawing or depositing) or to wait and keep their money balances unchanged. In the payment phase, the transaction size s realizes, and so does the acceptance policy of the shop visited by the agent. The function EV represents the expected value of the problem for an agent with money holdings m in the payment phase is given by

$$EV(m) = -Rm + \phi \int_0^{+\infty} V^a(m, s) dF(s) + (1 - \phi) \int_0^{+\infty} V^n(m, s) dF(s),$$

where $\mathcal{S}$ is the support of $F(\cdot)$ and $V^a(m, s)$ ($V^n(m, s)$) respectively represent the value of facing expenditure size s when cash on hand is m in a store where cashless payments are (not) accepted. The values of being in a store that accept cashless payments and in a store that does not are respectively

$$V^a(m, s) = \begin{cases} u(s) + \max \{-\kappa + \beta V(m), \beta V(m - s)\} & \text{if } s \leq m, \\ u(s) - \kappa + \beta V(m) & \text{if } s > m, \end{cases} \quad (3)$$

and by

$$V^n(m, s) = \begin{cases} u(s) + \beta V(m - s) & \text{if } s \leq m, \\ \beta V(m) & \text{if } s > m, \end{cases} \quad (4)$$

[Equation 3](#) says that when $s \leq m$ in a store that accept cashless payments, the agent has a payment method choice. When $s > m$, instead, paying cashless is the only viable alternative. [Equation 4](#) says that when $s \leq m$ in a store that does not accept cashless payments, the agent will surely pay in cash; when $s > m$ the agent will not be able to purchase the good and won't get utility from consumption.

3.2 Optimal decision rules

I now examine the properties of optimal decision rules in this setup. Unfortunately, the presence of a double discrete choice (an adjustment decision *and* a payment instrument choice) and of multiple state variables make the problem analytically intractable: I

am unable to obtain closed-form decision rules. Nonetheless, it is possible to obtain insights on the features of optimal policies under certain assumptions, and to verify that the assumptions hold for a wide range of parameters. This is what I do in this Section.

3.2.1 Withdrawal policies

In this one-sided inventory problem, the cash management policy can be summarized by the policy function for withdrawal choices. In the pre-payment phase, the agent can either adjust her money balances to the optimal value by withdrawing cash or keep them at their value and wait. The value of adjusting is independent of m , as there are no proportional adjustment cost but only the fixed cost η . Let $V^* = \max_{\hat{m}} EV(\hat{m})$. Then, by (2), the adjustment problem can be rewritten as

$$V(m) = \max \{V^* - \eta, EV(m)\}.$$

Definition 1. *The withdrawal policy function $(a(m), w(m))$ is given by an adjustment rule*

$$a(m) = \mathbb{1}(V^* + \eta < EV(m)),$$

and by a withdrawal size rule

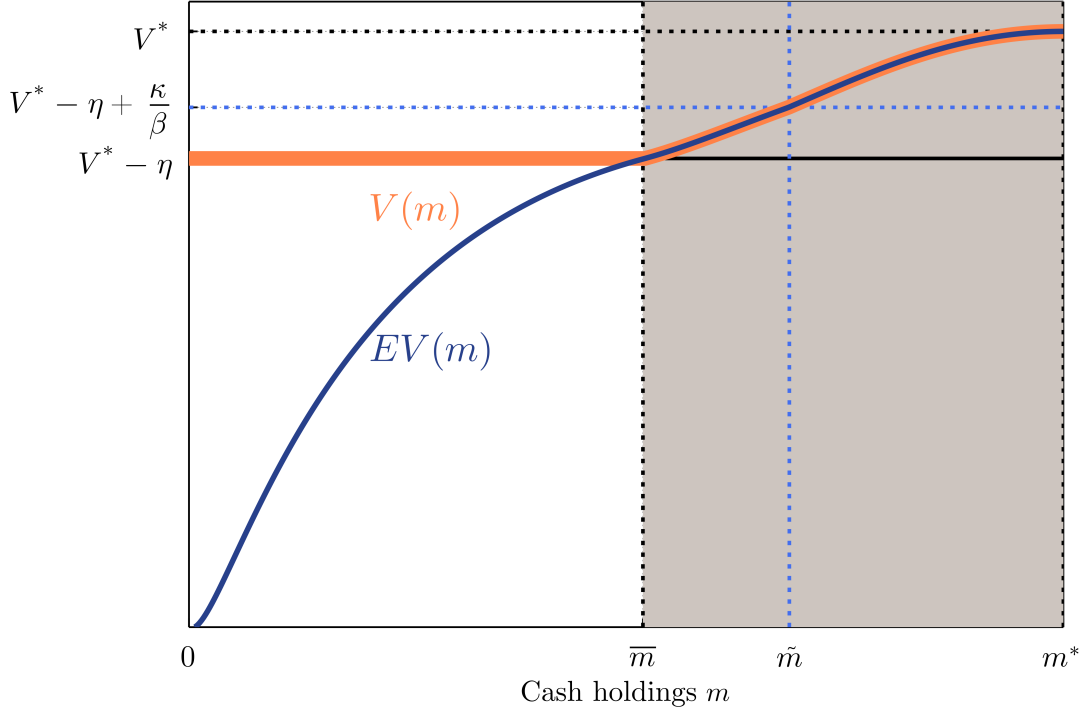
$$w(m) = \tilde{w}(m)a(m)$$

so that $a(m) = 1$ represents the choice to adjust cash balances through a withdrawal and $w(m)$ gives the withdrawal size, with $w(m) = 0$ if $a(m) = 0$.

For any reasonable parametrization, the optimal adjustment rules is of the form that can be observed in Figure 8. The Figure shows the value of adjusting $V^* - \eta$, the value of non-adjusting $EV(m)$ and their upper envelope $V(m)$. For the agent, it is not optimal to adjust cash holdings for levels of money balances equal or higher than a level $\bar{m}$; when $m < \bar{m}$, agents find it optimal to withdraw money. Of course, they will withdraw enough cash to reach an optimal level m^* ¹². Notice that in the Figure, the cash holdings level $\tilde{m}$ for which $V(m) = V^* - \eta + \kappa/\beta$ is also displayed. The next Subsection, which discusses payment choice policies in the model, will make clear why this level of cash

¹²Readers familiar with the inventory-theoretical literature will immediately recognize that the adjustment policy is a classical (s, S) -rule. These decision rules are optimal in a wide class of problems with fixed adjustment costs, such as inventory management problems (Scarf (1959), Iglehart (1963), Scarf (2005)). If they follow an (s, S) -rule, agents do not replenish inventories until their level falls below s ; when it does so, they make an order and adjust up to the level S . My cash management problem is extremely similar to a periodic-review inventory one: instead of placing orders, people withdraw cash from ATMs. As in standard discrete-time inventory models, a necessary and sufficient condition for this kind of policy to be optimal is η -concavity of $EV(m)$ (see Hall and Rust (2007)). In this paper, we cannot prove that $EV(m)$ has this property for any parameter configuration, but we can easily check if for a given parameter vector the property holds. All our results assume η -concavity and continuity of $EV(m)$, which turn out to be satisfied for any calibrated transaction size distribution and for all parameter vectors we estimate in Section 4.

FIGURE 8: Cash holdings adjustment policy function.



holdings is relevant. Let's now formally describe the optimal withdrawal policies.

Proposition 1 (Withdrawal policies). *Let*

$$\begin{aligned} m^* &= \inf \arg \max_{\hat{m}} EV(\hat{m}) \\ \bar{m} &= \inf \left\{ m \in [0, m^*] \mid EV(m) \geq EV(m^*) - \eta \right\} \end{aligned}$$

The withdrawal policy function is given by the adjustment and withdrawal rules

$$\begin{aligned} a(m) &= \mathbb{1}(m < \bar{m}), \\ w(m) &= (m^* - m) \mathbb{1}(m < \bar{m}), \end{aligned}$$

and it is possible to express the value of the problem as

$$V(m) = \begin{cases} V^* - \eta & \text{if } m \in [0, \bar{m}), \\ EV(m) & \text{if } m \in [\bar{m}, m^*], \end{cases}$$

where $V^* = EV(m^*)$.

Proof. The proof can be simply adapted from [Hall and Rust \(2007\)](#). However, it is a well-known result in inventory theory. ■

Since there are no exogenous inflows of cash in the model, m will never be higher than

m^* . People will withdraw money and reach m^* , and they will manage their transactions optimally using cash or cashless means of payments until their cash balances reach (or go below) $\bar{m}$: then, the cycle will start again. The region $[\bar{m}, m^*]$ is sometimes defined as the *inaction region*.

3.3 Payment choice policies

The true novelty of the model is the presence of a payment choice policy. Payment choice policy functions are two-dimensional, as they depend on cash holdings m and on the size of the transaction s . Recall that agents really have a nontrivial payment choice when they are in a shop that accepts cashless payments ($j = a$) and when their cash in advance constraint does not bind ($s \leq m$). I call transactions that satisfy both conditions as *unconstrained payments*. The value of the problem for an agent facing an unconstrained payment of size s when holding m units of cash is given by

$$V^{pay}(m, s) = \begin{cases} \max \{ \beta V(m) - \kappa, \beta V(m - s) \} & \text{if } s \leq m - \bar{m}, \\ \max \{ \beta V(m) - \kappa, \beta (V^* - \eta) \} & \text{if } s > m - \bar{m}. \end{cases}$$

Indeed, when the incoming transaction is small enough that it would not push the agent in the withdrawal region $[0, \bar{m})$, the choice depend on s as well as on m , because the continuation value depends on $m - s$. On the contrary, when the incoming payment is big enough that it would lead to a withdrawal in the next period (as future cash holdings in the case of no withdrawals would be smaller than $\bar{m}$), the choice does not depend on s but just on the current cash holdings level m .

Definition 2. Let the payment choice policy function be given by

$$p(m, s) = \begin{cases} \mathbb{1}(\beta V(m) - \kappa > \beta V(m - s)) & \text{if } s \leq m - \bar{m}, \\ \mathbb{1}(\beta V(m) - \kappa > \beta (V^* - \eta)) & \text{if } s > m - \bar{m}. \end{cases}$$

so that $p(m, s) = 1$ when it is optimal to pay using cashless methods.

The next Proposition describes the main features of optimal payment choice decision rules.

Proposition 2. (Payment choice policies) If $0 < \kappa < \beta\eta$, then $\exists \tilde{m} \in (\bar{m}, m^*)$, implicitly given by,

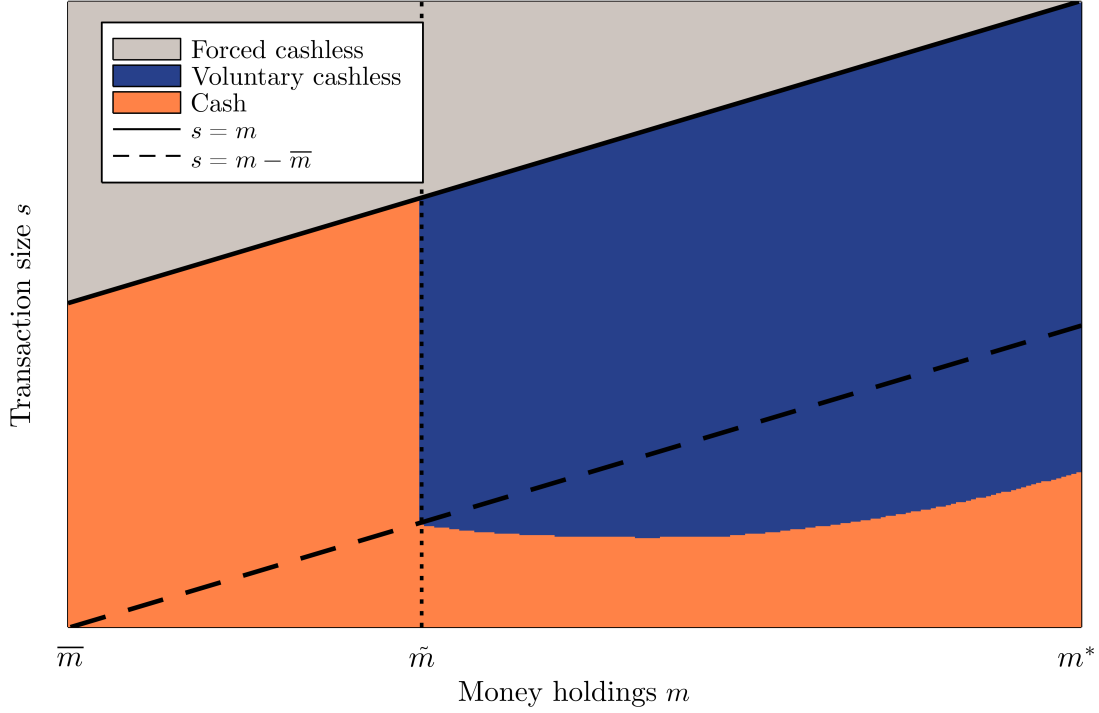
$$V(\tilde{m}) = V^* - \eta + \frac{\kappa}{\beta}, \quad (5)$$

such that

1. for $m \in (\bar{m}, \tilde{m}]$, for all $s \leq m$

$$p(m, s) = 0.$$

FIGURE 9: Payment choice policy function.



Note: Parameters are set to their baseline values. The model is solved numerically using value function iteration on a two-dimensional logarithmic grid for (m, s) and assuming a lognormal distribution $f(s)$ for transaction sizes. The solid black line represent the (m, s) -locus where the cash in advance constraint is binding ($m = s$): above it, there cannot be cash payments. The dashed black line represent the (m, s) -locus where paying by cash leads to the target level $\bar{m}$ ($m = s + \bar{m}$).

2. for any $m \in (\tilde{m}, m^*]$ there exists a nonempty set of transaction sizes $S(m) = (\underline{s}(m), m)$ such that

$$p(m, s) = 1, \quad \forall s \in S(m).$$

Moreover, $\bar{s}(m) = m - \bar{m} \in S(m)$ for all $m > \tilde{m}$.

If $\kappa > \beta\eta$, then $p(m, s) = 0$ for all m and $s \leq m$.

Proof. See [Appendix A.2](#). ■

The above Proposition says that, if the cost of using the card is lower than the discounted cost of withdrawals, there exist a level of cash holdings $\tilde{m}$ such that, when $m \in (\bar{m}, \tilde{m}]$, it is always optimal to pay using cash. Notice that $\tilde{m}$ is the highest level of cash holdings for which agents prefer to pay cash for transactions that would push them in the withdrawal region $[0, \bar{m})$. In this region, cash holdings are so far from the optimal level m^* that it's not worth it to pay κ in order to keep them intact. When $m > \tilde{m}$ and $s > m - \bar{m}$, instead, cash holdings are so close to the optimal ones that agents do not want to deplete them and immediately be forced to withdraw, therefore they use their cards. Finally, when $m > \tilde{m}$ and $s < m - \bar{m}$, payment choice decisions

depend on the transaction size s as well. Indeed, for any given m in this region, it can be optimal to pay using cashless methods in order to preserve money holdings, but if and only if the incoming transaction s is sufficiently high. This happens because the agent compares the extra cost of a cashless payments κ with the difference in expected costs $\Delta V(m, s) = \beta (V(m) - V(m - s)) < 0$: she decides to pay using cashless only if the reduction in expected future cash management cost is high enough to compensate for the extra transaction cost κ . Of course, when s is very small, $\Delta V(m, s) \simeq 0$, so it's never worth it to pay κ , as the future cash management benefits are minimal. As Figure 9 shows, when s is higher than a certain threshold $\underline{s}(m)$ using cashless is optimal, in order to keep cash balances intact: in this cases, paying using cash would decrease money holdings to a level so low that the expected discounted cost would be much higher; to avoid it, agents are willing to pay κ . From now on, I call these payment choice policies (m, s) policies.

3.3.1 Relationship with previous work

I now highlight that some of the features of decision rules found by Whitesell (1989) and Alvarez and Lippi (2017) are also present in this model, i.e., that (m, s) policies can be thought of as a more general decision rules which include features of payment choice policies discussed in previous work on the topic..

A rationalization of transaction-size thresholds. As said before, the observed negative correlation between transaction sizes and frequency of cash usage motivated Whitesell (1989) to derive transaction-size threshold policy rules for payment choices: when the size of a payment is larger than the threshold $\bar{s}$, in his model, cashless payments are optimal. However, he derived his result under two assumptions: i) agents minimize the steady-state cost of policies, with no explicit dynamic optimization or uncertainty involved¹³; ii) the cost of using cashless methods depends on the size of the transaction, i.e., it has form $\kappa_0 + \kappa_1 s$. In the model presented in this paper, there is no need to assume that the cost of cashless payments depends on s to obtain such rules, or to arbitrarily transform an inherently dynamic stochastic problem such as sequential payment choices in a deterministic and static one. Transaction size threshold policies emerge naturally as part of (m, s) policies from the problem's dynamic structure, which makes agents internalize the consequences of their payment choices on expected cash management costs. For any level of cash holdings $m > \tilde{m}$, there exists an m -specific transaction size threshold $\underline{s}(m)$ and agents find it optimal to use cashless payments if

¹³Whitesell (1989) assumes that albeit transaction sizes are not all equal, the total amount spent in each period is fixed and so is the fraction of incoming transactions of each size. Therefore, in his setup agents just choose the total amount of cash to hold, and they never run out because expenditures can never exceed their expected value: the model is deterministic. It is also static, i.e., the evolution of m in response to different sequences of transaction sizes in a given period is not modeled: given full certainty, it is in fact irrelevant, as the average cash balances in all periods (and therefore opportunity costs) will be the same independently of the sequence in which expenditures come. This creates independence of decision rules with respect to m , as decisions are taken at the start of the period

$s > \underline{s}(m)$. This can be seen as a generalization of the theory based on the existence of a unique transaction size threshold: thresholds are not only individual specific (as they depend on policy parameters which are clearly heterogeneous, such as κ or R), they are transaction-specific, as they depend on cash holdings.

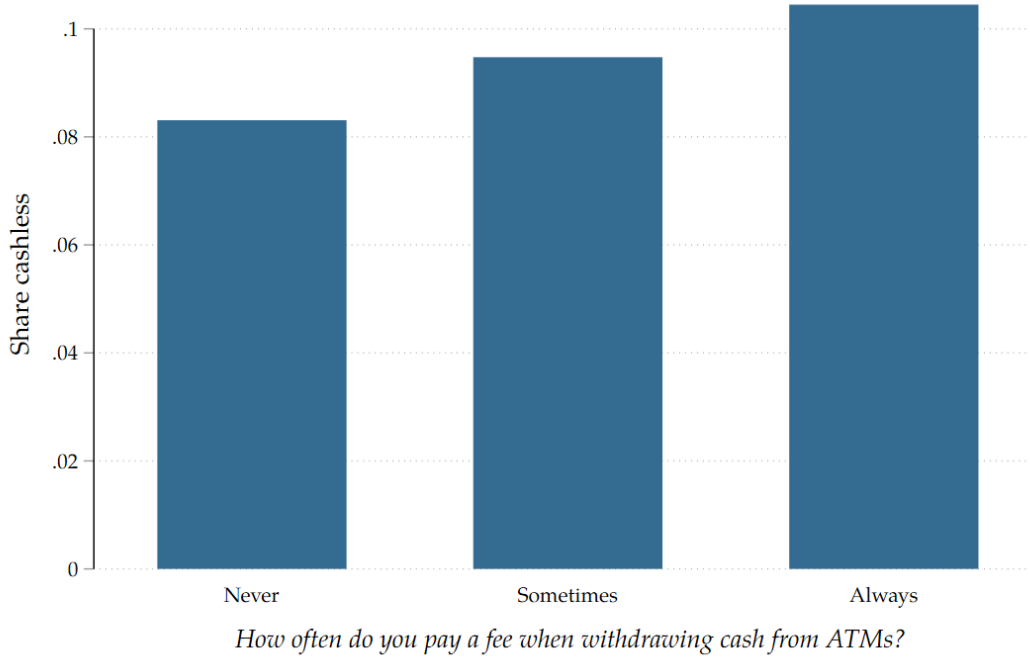
A rationalization of cash burns policies. As said above, optimal policies in [Alvarez and Lippi \(2017\)](#) are the so-called *cash burns* policies. People pay in cash whenever they have enough (which in their setting means any time agents have $m > 0$, as the consumption stream is infinitesimal). (m, s) policies have a similar property: when s is lower than some threshold, it is never optimal to settle transactions using a cashless method. This is because the gain of using the card (avoiding a depletion in cash balances) is so low (as s is small, cash balances are relatively unaffected) that it's not worth it to pay the fixed cost κ . When s is extremely small, as in [Alvarez and Lippi \(2017\)](#) it is perfectly reasonable that agents prefer to use cash, as they do not push money holdings down and at the same time, they pay no cost.

3.4 Model predictions and the data

A novel result that emerges from optimal payment choice policies is that cashless payments can be optimal even if they are a more expensive payment method than cash ($\kappa > 0$), as long as they are a cost-effective cash management tool. This happens to be the case if and only if $\kappa < \beta\eta$, i.e., if it is more convenient to pay κ at the time of the payment to keep money balances intact with respect to paying with cash and then (in the next period) withdrawing. To see why it's the case, think of the opposite scenario, in which $\kappa > \beta\eta$: for any (m, s) , it's never convenient to pay cashless. If one wants to keep money balances at their level, it's cheaper to pay using cash, then withdraw to restore the previous level of cash holdings. Therefore, the model has a sharp prediction: voluntary cashless usage by agents who dislike using cards is possible only when the fixed cost of a cashless payment is lower than the discounted fixed cost of an ATM withdrawal.

This prediction is not easily testable as neither κ nor η are observed in the data. However, if one is willing to make some assumption about the relationship of κ and η with observables, it is possible to obtain some insights. As our measure of κ we can exploit the answer to the question about preferred payment methods. Let's focus on agents that say they prefer cash, so that $\kappa > 0$ (as in the model) seems a reasonable assumption. Ideally, one would like to keep κ fixed and see how the share of voluntary cashless payments changes with η . As a proxy for η , we can exploit the answer to the question "How often do you have to pay fees when withdrawing at ATMs?", which was asked in the SUCH survey (2016). Agents report if they never pay fees when withdrawing, if they sometimes have to pay or if ATM usage always entails the payment of a fee. In [Figure 10](#) I display the share of voluntary payments which is performed using cashless methods

FIGURE 10: Withdrawal costs and voluntary cashless payments.

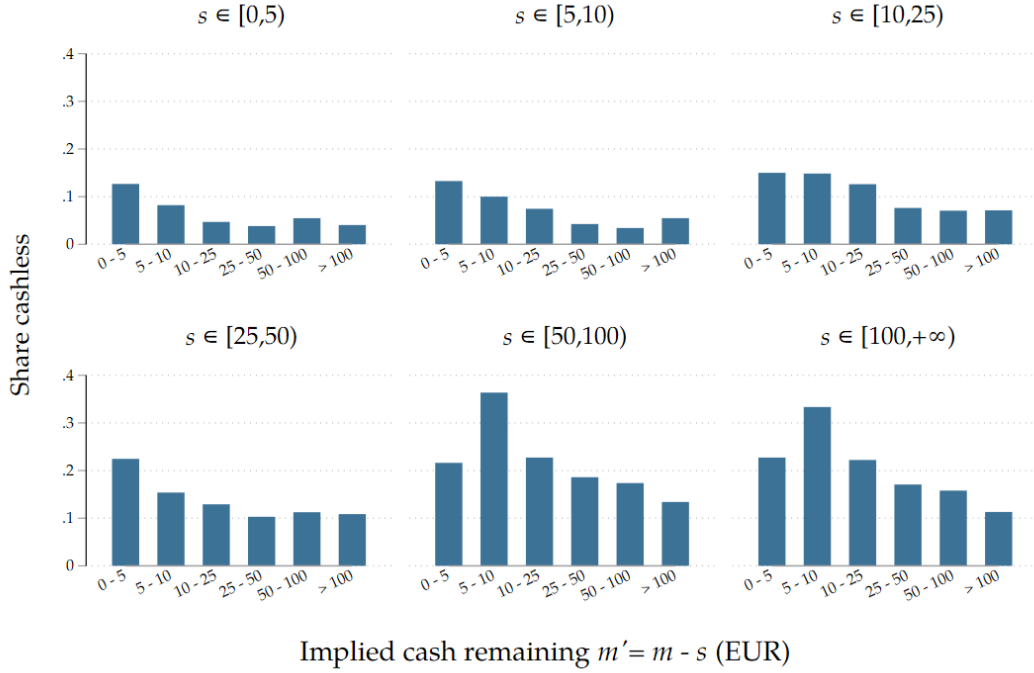


Note: Information on usual withdrawal costs faced by consumers is obtained from answers to the question “How often do you have to pay fees when withdrawing at ATMs?”. An higher likelihood of paying fees when withdrawing is a proxy of higher withdrawal costs η . I only consider voluntary payments, and I restrict the sample to people who reportedly prefer to use cash, to keep κ fixed while η varies. Source: ECB SUCH Data (2016).

as a function of the proxy for η . Results show that higher withdrawal costs are associated with a higher share of voluntary payments being settled using cashless methods, which is consistent with the theoretical implications of the model regarding the cash management role of card payments. When agents find it harder to withdraw money, they increase card usage to preserve higher cash balances, instead of using them and replenish at ATMs more often.

Another prediction of the model is the following: in order to minimize cash management costs, it is more convenient to perform voluntary cashless payments when i) money holdings are close to their optimal level m^* , and ii) an incoming transaction s would significantly deplete their cash holdings, i.e., $m' = m - s \ll m^*$. It is possible to test using SUCH and SPACE data if this is really the case. As before, I focus on agents that reportedly prefer to use cash, to keep κ fixed, and I investigate whether they are more likely to perform cashless payment when implied future cash holdings in the case of a cash payment ($m' = m - s$) are low than when they are at an intermediate level. This would be consistent with the model, as agents want to avoid holding too little cash and they are willing to pay cashless (even though they dislike it with respect to paying cash) to keep their money balances high. In [Figure 11](#) I show that the data

FIGURE 11: Voluntary cashless usage by cash lovers and implied cash remaining.



Note: This graph is based on transactions performed by people who reportedly prefer cash, in which both payment options were available. It documents the frequency of voluntary cashless usage by cash lovers for different values of $m' = m - s$, the level of cash remaining if the transaction is settled using cash, and for different levels of s (the size of the transaction). Source: ECB SUCH (2016) and SPACE (2019) Data.

seem to confirm this hypothesis. The Figure shows that agents that prefer to use cash are more willing to settle their transaction using a card if implied cash holdings are very low, while the share of voluntary cashless transaction they make is much lower when implied cash holdings are at an intermediate or high level. This holds for any level of transaction sizes s , despite the intensity of cashless usage is way higher when s increases. It also makes sense that the effect of a decrease in m' on payment choices is more pronounced for high s : when s is small, the gain in future expected cash management costs associated with card usage is lower, as m and m' are very close; when s is large, m and m' are significantly different and therefore there is more scope for card usage, i.e., it is more likely that the benefits of using the card (lower expected cash management costs in the future) exceed its direct cost κ .

4 Explaining Cross-Country Differences

I now turn to assessing the model's potential to replicate relevant features of the data. The purpose of this final part of the paper is not to formally establish conditions under which identification of the model can be achieved, nor to perform a quantitative estimation exercise that would allow to take seriously the estimated model and to use

TABLE 3: Cash management and payment choices, cross-country evidence.

	<i>Cash management</i>				<i>Payment choices</i>		
	M	W	$\underline{M}$	ϕ	$\frac{n_{cless}}{n}$	$\frac{n_{cless}^{vol}}{n^{vol}}$	$\frac{exp_{cless}}{exp}$
Austria	60.0	50.0	17.4	0.71	0.15	0.25	0.31
Belgium	40.2	40.0	19.0	0.73	0.32	0.44	0.62
Estonia	21.4	30.0	7.0	0.83	0.50	0.59	0.67
Finland	31.2	40.0	14.5	0.85	0.46	0.51	0.66
France	24.3	20.0	11.4	0.82	0.30	0.32	0.69
Greece	45.2	50.0	31.0	0.57	0.13	0.28	0.28
Ireland	44.5	50.0	10.6	0.72	0.18	0.25	0.34
Italy	50.0	50.0	30.0	0.66	0.13	0.25	0.30
Latvia	24.2	20.0	9.0	0.68	0.30	0.45	0.50
Lithuania	30.6	40.0	12.3	0.74	0.24	0.39	0.41
Luxembourg	68.6	80.0	18.7	0.80	0.36	0.43	0.66
Portugal	20.1	20.0	7.7	0.67	0.17	0.25	0.45
Slovakia	30.0	30.0	15.2	0.58	0.22	0.46	0.35
Slovenia	30.7	30.0	17.5	0.79	0.20	0.30	0.38
Spain	32.5	25.1	20.0	0.66	0.12	0.22	0.32
Total	36.9	38.3	16.1	0.72	0.25	0.36	0.46

Note: The first three moments are related to cash management. M are median cash holdings (EUR), W is the median withdrawal, $\underline{M}$ is the median cash balance at the time of a withdrawal. The last four moments concern payment choices: ϕ is the level of card acceptance in the country, n_{cless}/n is the share of cashless payments, n_{cless}^{vol}/n^{vol} is the share of voluntary payments settled using cashless methods and exp_{cless}/exp is the cashless share of total expenditure. Source: SUCH (2016) Data.

it for policy analysis. My scope here is to show that the model can replicate some key data moments concerning payment choice and cash management, and to show that sizable heterogeneity in parameters is needed to match the cross-country difference in cash management and payment behavior.

4.1 Payment and cash management behavior in the Euro Area

Before calibrating the model, I will now present data showing what is the extent of heterogeneity in payment and cash management behavior in the Euro Area. In Table 3 I display seven relevant moments on cash management and payment choices for all countries in the data in 2016, using SUCH data¹⁴. In particular, regarding cash management behavior, I display median cash holdings M , the median withdrawal W and the median level of cash on hand when withdrawing $\underline{M}$. As for payment choices, I fo-

¹⁴I use SUCH instead of the most recent SPACE survey because the information on withdrawals is more accurate and leads to more precise computation of W and $\underline{M}$.

cus on card acceptance levels ϕ , on the share of cashless payment n_{cless}/n , on the share of voluntary payments settled using cashless methods n_{cless}^{vol}/n^{vol} and on the cashless share of total expenditure exp_{cless}/exp .

As the Table shows, there is huge heterogeneity across the Euro Area along all these dimensions. Median cash balances held by consumers range from around 20 euros (as in Portugal and France) to more than 60 euros (as in Austria or Luxembourg). In some countries, including Austria, Greece, Ireland and Italy, the median withdrawal is as high as 50 euros; people from France, Latvia and Portugal, instead, withdraw relatively little cash, with median withdrawals around 20 euros. The median Italian or Greek consumer goes to withdraw money at an ATM when she still has 30 euros on hand (2/3 of her median cash balances); in France and Portugal, consumers visit ATMs when they have approximately 10 euros left (one third of their median balances). Cashless acceptance is highly heterogeneous as well: it ranges from almost perfect (as in Finland) to very low (Austria, Italy and especially Slovakia). The same holds for the share of payments settled using cards, as well as for the share of voluntary payments in which cashless methods are used and for the cashless share of expenditure.

4.2 Estimation strategy

I solve the model by discretizing the state space on a two-dimensional grid $\mathcal{M} \times \mathcal{S}$, where $\mathcal{M}$ is the vector of possible cash holdings and $\mathcal{S}$ is the vector of possible transaction sizes. I pick $\max \mathcal{M} = \max \mathcal{S} = 250$ because more than 99.5% of transactions are characterized both by $m < 250$ and by $s < 250$. I also pick $\min \mathcal{M} = 0$ since agents cannot have negative cash holdings and $\min \mathcal{S} = 0.01$ as the smallest payment observed in the data has this size. I set the number of grid points equal to 200 for each dimension (after checking that results are invariant to increasing precision) and find the solution through standard value function iteration, obtaining value functions $V(m)$, $EV(m)$, $V^a(m, s)$ and $V^n(m, s)$, as well as policy functions $a(m)$ and $p(m, s)$.

Some parameters of the model can be calibrated directly from SUCH data. First, regarding the transaction size distribution $F(s)$, I assume that $s \sim LN(\mu^s, \sigma^s)$, with μ_s and σ_s chosen to match the mean and standard deviation of the observed distribution of payments in each country¹⁵. Second, concerning the acceptance level ϕ , I simply compute it from the data by looking at the share of payments for which cashless usage was possible. For any country i , I have a set of calibrated parameters $\Gamma_i = (\mu_i^s, \sigma_i^s, \phi_i)$. Finally, to keep things simple, I assume that the utility derived from settling transactions is a linear function of the size of the transaction, i.e., that $u(s) = s$.

I estimate the remaining four parameters of the model $\Theta = (\beta, \eta, R, \kappa)$ via a simple simulated minimum distance procedure. Note that for any set of parameters Θ , af-

¹⁵The choice of a lognormal distribution is based on Boeschoten (1992) and on SUCH data showing that a lognormal specification nicely fits the observed distribution of transaction sizes. See Appendix A.3.

ter solving the model and getting the policy functions, I can compute the stationary distribution of cash holdings $\Lambda(m)$ through a simple recursive algorithm. Then, I can initialize a cross section of $N = 100000$ agents drawing their initial cash holdings from $\Lambda(m)$ and simulate their cash management and payment choices, after randomly assigning to each of them a payment size and a type of store. With this simulated data at hand, I can easily compute four moments: median cash holdings M , median withdrawal W , median cash-at-withdrawal $\bar{M}$ and the share of voluntary payments settled in cash n_{class}^{vol}/n^{vol} . For any country i , the vector of estimated parameters $\hat{\Theta}_i$ is given by

$$\hat{\Theta}_i = \arg \min_{\Theta} \Psi_i(\Theta) = \arg \min_{\Theta} \mathbf{e}_i(\Theta)' I_{4 \times 4} \mathbf{e}_i(\Theta), \quad (6)$$

where, given some simulated moment $\mu_{i,k}^s(\Theta)$ and the corresponding data counterpart $\mu_{i,k}^d$, we have

$$\mathbf{e}_i(\Theta) = \left(\frac{\mu_{i,k}^s(\Theta) - \mu_{i,k}^d}{\mu_{i,k}^d} \right)_{k=1}^4,$$

i.e., moment deviations are rescaled to avoid unintended weighting due to different units. To find the minimizer of $\Psi_i(\Theta)$, I exploit a multi-start global optimization method sharpened with a final local optimization step to make sure that the minima I find do not depend on the arbitrary initial guesses.

4.3 Estimation results

In [Table 4](#) I display the estimation results. Despite these estimated parameters are hard to interpret quantitatively, their relative magnitude and cross-country comparison can still be used to shed light on the quantitative differences in tastes (β, κ) , technology (η) and environmental variables (R) that cause such big differences across European countries. The heterogeneity in parameters needed to reproduce the observed differences in payment and cash management behavior are noteworthy. In order to fit high cash holdings and big withdrawals in countries such as Austria, Ireland and Italy, for instance, one needs big withdrawal costs η and a pretty high fixed cost of using the card κ . French, Finnish and Portuguese people seem to have different tastes on payment methods (a much lower κ) and also to have easier access to the withdrawal technology (lower η).

Note that, for all countries, the estimated κ is positive. This could lead to the conclusion that cash is still the preferred method in every country. However, I need to make clear that the absolute magnitude of κ in the present model should not be taken too seriously, and the reader should only focus on relative comparisons between countries. Indeed, a positive value of κ is needed to avoid that the share of voluntary payments settled using cards n_{class}^{vol}/n^{vol} equals one. To see why, think of a country in which everyone prefers using the card ($\kappa < 0$). In that environment, people would bring cash with them just

TABLE 4: Estimated parameter values and estimated minimum distance between theoretical and empirical moments for all countries.

	$\hat{\beta}$	$\hat{\eta}$	$\hat{R}$	$\hat{\kappa}$	$\Psi(\hat{\theta})$
Austria	0.993	3.073	0.056	0.956	0.033
Belgium	0.995	1.38	0.059	0.35	0.015
Estonia	0.955	1.832	0.041	0.232	0.017
Finland	0.944	1.288	0.033	0.255	0.025
France	0.977	0.504	0.042	0.18	0.013
Greece	0.982	1.962	0.055	0.548	0.074
Ireland	0.976	3.089	0.061	0.98	0.014
Italy	0.996	1.798	0.046	0.568	0.026
Latvia	0.975	0.808	0.069	0.199	0.015
Lithuania	0.973	2.262	0.046	0.422	0.01
Luxembourg	0.993	3.918	0.036	0.645	0.005
Portugal	0.995	0.817	0.069	0.316	0.015
Slovakia	0.942	1.411	0.058	0.222	0.014
Slovenia	0.961	0.824	0.049	0.299	0.027
Spain	0.971	0.542	0.048	0.231	0.011

to avoid situations in which cards are not accepted: they would never use it to settle transactions in stores where cards are accepted, as this would entail both a static and a dynamic loss (they would use their least prefer payment method and remain with less precautionary cash balances). In a more complete model, there should ideally be a distribution of κ instead of a unique value for each country: that could lead to realistic behavior even allowing $\mathbb{E}(\kappa)$ to be negative, i.e., cashless methods to be preferred on average.

The last columns of Table 4 summarizes how well does the model fit the data, i.e., it displays the distance between the true moments and the simulated ones computed by solving the model given the estimated parameters. More detailed information on how closely the model is able to replicate real payment and cash management behavior is provided by Table 5, that shows the fit of the model for each country in the sample, comparing simulated moments derived from the estimated models with their data counterparts.

4.4 Possible extensions: a quantitative model

The purpose of this paper is showing that a relatively simple model that includes heterogeneous transaction sizes into a dynamic cash management framework with a payment method choice is able to generate realistic features of payment behavior, such as

TABLE 5: Model fit for estimated parameter values $\hat{\theta}$.

		M	W	$\underline{M}$	$\frac{n_{cless}^{vol}}{n^{vol}}$
Austria	<i>Model</i>	50.34	55.37	18.46	0.27
	<i>Data</i>	60.0	50.0	17.4	0.25
Belgium	<i>Model</i>	40.27	36.91	16.78	0.44
	<i>Data</i>	40.2	40.0	19.0	0.44
Estonia	<i>Model</i>	23.49	31.88	6.71	0.61
	<i>Data</i>	21.42	30.0	7.0	0.59
Finland	<i>Model</i>	35.23	41.95	11.74	0.53
	<i>Data</i>	31.18	40.0	14.5	0.51
France	<i>Model</i>	25.17	23.49	11.74	0.32
	<i>Data</i>	24.3	20.0	11.38	0.32
Greece	<i>Model</i>	53.69	45.3	25.17	0.3
	<i>Data</i>	45.23	50.0	31.0	0.28
Ireland	<i>Model</i>	41.95	48.66	13.42	0.25
	<i>Data</i>	44.5	50.0	10.63	0.25
Italy	<i>Model</i>	57.05	46.98	26.85	0.25
	<i>Data</i>	50.0	50.0	30.0	0.25
Latvia	<i>Model</i>	21.81	20.13	8.39	0.46
	<i>Data</i>	24.19	20.0	9.04	0.45
Lithuania	<i>Model</i>	31.88	36.91	11.74	0.39
	<i>Data</i>	30.57	40.0	12.32	0.39
Luxembourg	<i>Model</i>	65.44	83.89	18.46	0.44
	<i>Data</i>	68.62	80.0	18.7	0.43
Portugal	<i>Model</i>	20.13	20.13	6.71	0.24
	<i>Data</i>	20.1	20.0	7.65	0.25
Slovakia	<i>Model</i>	33.56	30.2	15.1	0.47
	<i>Data</i>	30.0	30.0	15.2	0.46
Slovenia	<i>Model</i>	33.56	28.52	15.1	0.31
	<i>Data</i>	30.7	30.0	17.46	0.3
Spain	<i>Model</i>	33.56	23.49	18.46	0.23
	<i>Data</i>	32.5	25.07	20.0	0.22

voluntary usage of cashless payments by agents that can have the option to pay using cash. The model presented is deliberately simple and it does not allow for a full quantitative analysis based on a more rigorous estimation procedure. The first issue with this framework is that decision rules are deterministic: depending on parameter values, for each (m, s) pair there is only one optimal action. This imposes restrictions which are not likely to hold in the data. Payment choice policy functions that only depend on

observables (m and s) would create a zero likelihood problem in a proper estimation exercise. A better, quantitative model would include choice-specific utility instead of the unique parameter κ to proxy for the relative costs of paying cash or cashless at POS. Utility could depend on s , for instance, so as to incorporate supply-side factors. Moreover, one would need some unobserved component, such as random utility shocks to put nonzero probabilities on each choice and to avoid zero likelihood problems.

In my setting, withdrawal policy functions are also deterministic, depending only on m . A way to avoid this would be to allow for random fixed costs, as proposed by [Alvarez and Lippi \(2017\)](#): in the data, agents do not only receive money by withdrawing at ATMs; sometimes they just get it from a cash jar at home or as cashback when shopping at a store. In fact, the data shows that around 15% of upward cash adjustments have these sources. Allowing for random variation in η (and estimating a distribution of fixed costs, instead of a unique value) would allow for more flexible adjustment rules. Moreover, in the model there is a unique target level of cash balances: after a withdrawal, $m = m^*$ in any case. In reality, however, post-withdrawal balances have a non-degenerate distribution, even at the individual level. A way to obtain a non-degenerate distribution would be to allow individuals to receive a signal about the size of the incoming payment when making their withdrawal choices: in this way, they would decide to withdraw more (and more often) when there is a big incoming transaction than when they think the next purchase will be a small one.

Another shortcoming of the model is that agents never deposit cash, as they cannot have more cash than what they desire. Albeit there is no specific information on deposits in SUCH/SPACE data, future payment diaries could incorporate this information, allowing for the estimation of more flexible adjustment rules. In an extended version of this model, one could allow for exogenous *inflows* of cash: in SUCH/SPACE data, I observe that around 25% of cash is obtained from family, friends, colleagues or as cash income. Including inflows of cash would however make the problem a two-sided inventory problem, and the associated policy function would be much trickier than those studied here, and maybe close to the standard adjustment rules of two-sided models like [Miller and Orr \(1966\)](#).

The inclusion of these features would allow one to take seriously the estimation procedure of parameters using tools from the dynamic discrete choice literature ([Rust \(1987\)](#), [Bajari, Benkard, and Levin \(2007\)](#)) or an indirect inference approach based on matching coefficients of logit models for adjustment and payment choices. A quantitative model of payment choice and cash management would be a key tool to study the consequences and welfare implications of measures such as limiting cash usage or increasing cashless acceptance rates via subsidies targeted to merchants. Moreover, it could be used to study the reasons behind the resilience of cash as a means of payment despite the existence of cheap substitutes, disentangling the effect of preferences for cash from that of imperfect card acceptance, and ultimately establishing the condi-

tions under which the transition to a cashless society could be achieved and whether it would be welfare-enhancing.

5 Conclusion

High-quality, transaction level data on payment method decisions of Euro Area consumers shows that existing theoretical models of payment choice fail to capture key facts of observed behavior. In this paper, I exploit evidence from European payment diaries to argue that a model of payment choices needs to: i) jointly study means of payment and cash management decisions; ii) take into account the fact that consumers' decisions on how to pay are affected *both* by the size of each transaction and by the level of cash on hand, and not only by one among these factors, as previously assumed in the literature; iii) avoid abstracting from imperfect acceptance of cashless methods. I present a novel framework to study cash management and payment choices, showing its theoretical properties and empirically verifying some of its predictions. The model account for observed choices that previous theories could not explain: in particular, it is the first to rationalize the fact that agents sometimes pay with cash even when they have enough cash holdings, and even if they state they prefer to use cash *ex-ante*: a key result is that this kind of behavior can arise as long as cashless payments are an effective cash management tool, i.e., they are "*cheaper*" than ATM withdrawals. I then show that an estimated version of the model can match cross-country heterogeneity in payment and cash management choices. I conclude by arguing that the model presented in the paper could be enhanced by the inclusion of features that would make it suitable for quantitative analysis. Future research on payment choices could make use of such a quantitative model to study the welfare effects of policies aimed at increasing cashless usage, such as limits to cash holdings or to cash payments, and to analyze the reasons behind the resilience of cash in modern advanced economies despite advancements in payment technologies.

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A Appendix

A.1 Data cleaning and payment choice sets

The raw version of the data made available by the ECB has a major shortcoming: participants are not asked to report the level of their cash holdings in the moments in which they perform transactions. The rationale behind this choice is not easy to understand, since cash holdings were included among the possible factors that shape the decision on how to pay in the questionnaire administered in the same survey.¹⁶ At a first glance, this lack of information may seem no big deal: given that agents reported their money balances at the start of the day, using the data on withdrawals and payments cash holdings should be straightforward to pin down at each point in time. Two issues arise nonetheless: first, a fraction of agents failed to report the timing of cash adjustments performed (in SPACE data no one did, as respondents were not asked); second, participants were not explicitly asked about possible cash deposits.

To minimize the loss of data related to these shortcomings, I adopted the following strategy. Concerning the first problem, I was able to recover the timing of cash replenishments for a share of the agents which failed to report it: using the fact that I can observe which payment method they employed in each transaction, I can pin down the timing of adjustments in all situations in which an adjustment is needed to explain a purchase (i.e., in which an agent purchased in cash a good worth more than her *unadjusted* cash holdings). All the individuals for which the timing of replenishments couldn't be pinned down were excluded from the data. The second issue can also be tackled exploiting the fact that agents reported their cash holdings at the end of the day. For all the agents for which we are able to track cash holdings, we can compute our predicted cash balances at the end of the day *assuming there were no deposits*. If the two figures differ, then an unreported deposit or replenishment must have taken place during the day, and we are therefore not sure that cash on end at time of each transaction is corresponding to the actual one. These observations were excluded as well, and so were the ones for which computed cash balances were negative at some point. Finally, all the observations for Malta and Cyprus were excluded as I found systematic differences between the reported cash at the end of the day and the one I computed¹⁷. The final sample I exploit for the analysis contains information on 176,593 transactions carried out by 100,471 individuals¹⁸. For a subsample I have access to a larger amount

¹⁶The absence of cash holdings could be due to the fact that previous literature (with some exceptions) did not stress their impact on payment decisions, focusing only on the size of the transaction. Further evidence about this is provided in the questionnaire, in which respondents were asked “*When shopping in shops, what is the amount below which you would typically pay with cash instead of other payment methods?*”. This question implicitly assumes that payment decisions are monotonic in transaction sizes, thereby ignoring the impact of cash on hand on choices.

¹⁷This does not happen with other countries. Maybe a coding error in the data provided by ECB?

¹⁸A share of participants actually did not carry out any payment in the day of analysis. The subsample of actual payers is composed by 75,187 individuals.

of information, since these participants also filled in the questionnaire. The variables I have access to in the cleaned version of the data can be divided in three families. First, I have access to *demographic data* on all participants. Second, I can observe a substantial amount of *transaction-level data* for all transactions performed. Third, for a subsample of participants (which includes a fraction of agents interviewed in the SUCH survey and all agents interviewed in the SPACE survey), *questionnaire data* are also available.

Payment choice set. A key aspect of the data is that it allows, for a large share of transactions, to perfectly observe the payment choice set of individuals, i.e., if they could use cash, cashless methods or both. First, tracking cash holdings at the moment of payments makes it possible to compare them with the associated transaction size and thereby to establish (for all 100,157 payments) if it was possible to carry out the transaction using cash: clearly, it is not possible to pay with cash if the transaction size is larger than current cash holdings. Formally, let m_{it} be cash holdings of individual i at time t and s_{it} be the transaction size she faces. Our dummy for the possibility to use cash $CashPoss_{it}$ is constructed by

$$CashPoss_{it} = \begin{cases} 1 & \text{if } s_{it} \leq m_{it} \\ 0 & \text{if } s_{it} > m_{it} \end{cases}$$

Second, combining information on card acceptance by merchants provided by respondents, payment methods employed and questionnaire answers, it has been possible to determine for a large share of payments if cashless payments were really an option for agents. There are two situations in which we are sure that cashless payments were an available option. The first situation, of course, is the one in which agents *did* use cashless methods to settle the transaction. The second situation is identified when two conditions are met: first, cashless payments must be accepted as a way to carry out the transaction (a condition I could check for all payments); second, the agent must have access to a payment card¹⁹ (which I could check for a large fraction of payments²⁰). Formally, let $CardOwn_i$ be a dummy equal to one if individual i owns a payment card (debit or credit) and to zero if she doesn't. Let $CashlessOwn_i$ be equal to one if individual i has access to at least a cashless payment method and equal to zero otherwise. Let $CashlessAcc_{it}$ be a dummy equal to one if for the payment faced at time t by individual i cashless payments are accepted. Let $Cash_{it}$ be a dummy equal to one if the payment method used by i at time t was effectively cash, and equal to zero otherwise.

¹⁹Since I don't know exactly which payment methods are accepted at any given transaction, but I just know if any cashless payments are accepted, I assume that an agent *could* have made a cashless payment at a store in which cashless instruments are accepted *if and only if* he/she has access either to a debit or to a credit card or both. The reason behind this is that credit and debit cards are always accepted in stores that accept some cashless payment method, while (for instance) cheques are not. Thereby, saying that an agent which has access to cheques (or credit transfers) could have paid cashless at a store in which cashless methods are accepted would be very risky, while for agents that own either a credit or a debit card it seems reasonably safe.

²⁰This are all the payments made by respondents to the questionnaire, for whom we have detailed information concerning access to cashless instruments, plus all the payments made by agents which performed at least a card payment.

Our dummy for the possibility to use cashless methods $CashlessPoss_{it}$ is constructed in the following way.

$$CashlessPoss_{it} = \begin{cases} 1 & \text{if } Cash_{it} = 0 \vee (CardOwn_i = 1 \wedge CashlessAcc_{it} = 1), \\ 0 & \text{if } CashlessAcc_{it} = 0 \vee CashlessOwn_i = 0, \\ \cdot & \text{otherwise,} \end{cases}$$

where $\cdot$ represent a missing value. For these transactions I can exactly pin down the payment choice set of individuals, and I'm thereby able to distinguish between *forced* (only cash or only payment card available) and *voluntary* (both available) payment choices. Formally, let Vol_{it} be a dummy variable equal to one when both payment options (cash and cashless) were available for individual i at time t and to zero when only one was available, which is constructed in the following way.

$$Vol_{it} = \begin{cases} 1 & \text{if } CashPos_{it} = 1 \wedge CashlessPos_{it} = 1 \\ 0 & \text{if } CashPos_{it} = 0 \vee CashlessPos_{it} = 0 \\ \cdot & \text{if } CashlessPos_{it} = \cdot \end{cases}$$

A.2 Proofs

Proof of Proposition 2

First, let's prove that $\exists \tilde{m}$ that satisfies (5). By the intermediate value theorem, given that V is continuous, that $V(\bar{m}) = V^* - \eta$ and that $V(m^*) = V^*$, $V(m)$ will take all the values in $(V^* - \eta, V^*)$. If $0 < \kappa < \beta\eta$, then $V^* - \eta + \frac{\kappa}{\beta} \in (V^* - \eta, V^*)$.

We now prove points 1. and 2. of the Proposition.

1. By contradiction. Suppose that $\exists s$ such that $p(m, s) = 1$ for some $m \leq \tilde{m}$, where $\tilde{m}$ is given by (5). This would mean that, for some $m \leq \tilde{m}$, and for some s ,

$$\beta V(m) - \kappa > \beta V(m - s) \geq \beta (V^* - \eta),$$

since $V(m) \geq V^* - \eta$ for all m . This implies that for some $m \leq \tilde{m}$,

$$\beta V(m) > \beta (V^* - \eta) + \kappa = \beta V(\tilde{m}),$$

which is impossible for $m \leq \tilde{m}$ (it cannot hold for $m = \tilde{m}$ and it contradicts the fact that V is non-decreasing in $[0, m^*]$ for $m < \tilde{m}$). Notice that for card usage to be optimal at $\tilde{m}$, one would need

$$\beta V(\tilde{m}) - \kappa = \beta (V^* - \eta) <> \beta V(\tilde{m} - s),$$

which clearly cannot hold.

2. Let $m > \tilde{m}$. First, I show that for all such m the set $S(m)$ is nonempty, i.e., that

there exist at least one payment size for each $m > \tilde{m}$ such that it is optimal to pay cashless. Simply pick $m = \tilde{m} + \varepsilon$, for some $\varepsilon > 0$. We proceed by contradiction. Assume that $\forall s$ it is optimal to pay using cash, i.e., that

$$\beta V(\tilde{m} + \varepsilon) - \kappa < \beta V(\tilde{m} + \varepsilon - s).$$

Pick $s = \tilde{m} + \varepsilon - \bar{m}$. Then, I get

$$\beta V(\tilde{m} + \varepsilon) - \kappa < \beta V(\bar{m}) = \beta(V^* - \eta).$$

Rearranging,

$$\beta V(\tilde{m} + \varepsilon) < \beta(V^* - \eta) + \kappa,$$

that can be rewritten as

$$\beta V(\tilde{m} + \varepsilon) < \beta V(\tilde{m}),$$

which clearly contradicts the fact that V is non-decreasing in $[0, m^*]$. Call $\bar{s}(m) = m - \bar{m}$, the transaction size that decreases current money holdings to the trigger value $\bar{m}$. We just showed that $\bar{s}(m) \in S(m)$ for all $m > \tilde{m}$.

We want to show that, for $m > \tilde{m}$, $\exists \underline{s}(m)$ such that $p(s, m) = 1$ for all $\underline{s}(m) < s \leq m$. For $m = \tilde{m} + \varepsilon$ and every $\varepsilon > 0$, $\underline{s}(m)$ is given by

$$\beta V(\tilde{m} + \varepsilon) - \kappa = \beta V(\tilde{m} + \varepsilon - \underline{s}(\tilde{m} + \varepsilon)).$$

Rearranging, we obtain

$$V(\tilde{m} + \varepsilon - \underline{s}(\tilde{m} + \varepsilon)) = V(\tilde{m} + \varepsilon) - \frac{\kappa}{\beta}.$$

Notice that $V(\tilde{m} + \varepsilon) - \frac{\kappa}{\beta} \geq V(\tilde{m}) - \frac{\kappa}{\beta} = V^* - \eta$. Again by the Intermediate Value Theorem, it certainly exists $\underline{s}(\tilde{m} + \varepsilon)$ that satisfies this Equation. It is clearly the case that for any $s > \underline{s}(\tilde{m} + \varepsilon)$,

$$V(\tilde{m} + \varepsilon - \underline{s}(\tilde{m} + \varepsilon)) < V(\tilde{m} + \varepsilon) - \frac{\kappa}{\beta},$$

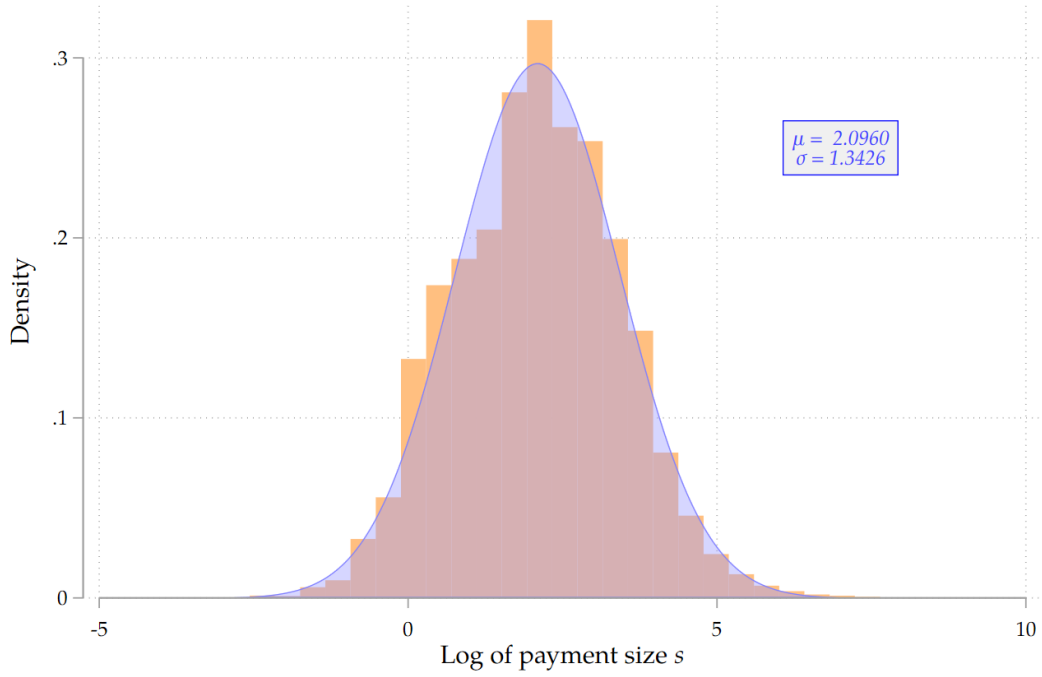
so that it is optimal to use cashless methods for all $s \in (\underline{s}(m), m)$.

A.3 The transaction size distribution $f(s)$

The transaction size distribution is a key object in the model, as it has a first order impact on the shape of the functions $V(m)$ and $EV(m)$, as shown in Appendix ??.

[Boeschoten \(1992\)](#) has noted that the distribution of transaction sizes is approximately lognormal. I investigate whether this is an appropriate assumption using all transactions in SUCH and SPACE data (more than 150,000 payments). As [Figure 12](#) shows, it

FIGURE 12: Distribution of transaction sizes.



Note: This graph is based all transactions in the data (155,546 payments). It displays the histogram of log payment sizes with an overlaid normal fit; estimated parameters of the normal distribution are also displayed. *Source: ECB SUCH (2016) and SPACE (2019) Data.*

turns out the $\log(s) \sim N(\mu, \sigma^2)$, therefore $s \sim LN(\mu, \sigma^2)$ appears to be an appropriate assumption. This is the specification we exploit to solve the model and obtain Figures and calibration results.